

NONBANK FINANCIAL INTERMEDIATION AND FINANCIAL STABILITY

A Perfect Storm in the Making?



GROUP OF THIRTY

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Foreword

The Group of Thirty is pleased to release *Nonbank Financial Intermediation: A Perfect Storm in the Making?*, which continues the Group of Thirty's (G30's) history of producing timely reports to address global challenges that impact the financial community and offer policy intervention considerations.

This study examines the rapid global growth of nonbank financial intermediation, considering both the opportunities presented and the risks that could amplify financial instability. The G30 report calls for targeted preventative reforms including stronger regulation, improved market infrastructure, enhanced data transparency, and coordinated international efforts to mitigate vulnerabilities. Planning for crisis, not just prevention, is

important too, calling for well designed liquidity facilities for nonbank institutions that stand a better chance of providing targeted support, in advance, rather than the open-ended interventions of the past.

On behalf of the Group of Thirty, I thank Project Co-Chairs, Agustín Carstens and Klaas Knot, for their dedicated leadership of the Working Group on NBFIs Reform, and Stijn Claessens for his careful drafting of the text. Additionally, I thank the G30 members who served as participants in the Steering Committee and Working Group for their time and their contributions to the project. The final report reflects the collective views of the Project Chairs and members of the Working Group, but does not reflect the views of the Group of Thirty as a whole.



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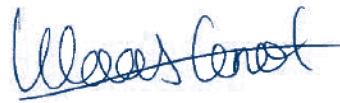
recommendations and overall analysis presented. On behalf of Stijn, we thank Viral Acharya, Nicola Cetorelli, Bill English, Susan McLaughlin, and Manmohan Singh for their very useful comments on an earlier version of the report.

The coordination of this project and many aspects of project management, Working Group logistics, and report production were centered at the G30 offices in Washington, D.C. This project could not have been completed without the efforts of our editor, Diane Stamm, and the work of the Group of Thirty team, Desiree Maruca and Maria Cueto Velez. We are grateful to them all.



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Abbreviations

AI	artificial intelligence
BCBS	Basel Committee on Banking Supervision
BoE	Bank of England
CCPs	central clearing counterparties
CNRF	Contingent Non-Bank Financial Institution Repo Facility
DeFi	Decentralized Finance
ESMA	European Securities and Markets Authority
EU	European Union
EUREP	Eurosystem repo facility for central banks
FIMA	US dollar repo facility for the official sector
FMI s	Financial Market Infrastructures
FSB	Financial Stability Board
FX	foreign exchange
G-SIB	Global Systemically Important Bank
GFC	global financial crisis
G7	Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States
HF s	hedge funds
HFT s	high-frequency traders
IC s	insurance corporations
IF s	investment funds
IMF	International Monetary Fund
MMF s	money market funds
NBFC s	nonbank financial corporations
NBFI	nonbank financial intermediation
NBFIs	nonbank financial intermediaries
PF s	pension funds

Executive Summary

Nonbank financial intermediation (NBFI) has grown rapidly worldwide since the 2008 global financial crisis (GFC).

By the end of 2024, NBFI assets reached \$260 trillion¹ globally, accounting for more than half of total financial intermediation. As central banks have scaled back sovereign bond purchases, NBFI has increasingly become a key source of government financing. It is also playing a growing role in financing the real economy, with private credit exceeding traditional bank lending to nonfinancial corporations in some jurisdictions. Beyond broadening access to finance and diversifying funding sources, NBFI offers important benefits, including stronger competition, improved pricing, enhanced financial services, and greater product innovation, with associated gains for sovereigns, businesses, and households alike. NBFI's risk-sharing structure and capacity to absorb shocks further supports its significant potential contribution to economic activity and resilience.

Its rapid expansion, however, raises new financial stability concerns.

The size and importance of NBFI—including its riskier segments—vary widely across jurisdictions. NBFI also encompasses a wide range of activities and entities,² many of which pose limited systemic risk, the focus of this report. Several NBFI types, nonetheless, exhibit structural vulnerabilities comparable to those in traditional banking. Many Nonbank Financial Intermediaries (NBFIs) combine illiquid assets with runnable liabilities, resulting in significant mismatches. For money market and other investment funds, large-scale redemptions during periods of stress can quickly generate liquidity shortfalls. Consolidating on- and off-balance

sheets, other NBFIs operate with very high leverage. In core bond markets, hedge funds and other globally active firms dominate trading, often holding large, highly leveraged, and short-term funded positions that can unwind abruptly following shocks.

Margin calls on derivatives and hedging positions can further amplify liquidity demands during periods of market volatility. Because banks are often unwilling to provide liquidity to NBFIs in stress periods, and NBFIs lack direct access to central bank facilities, liquidity shortfalls can spill over to the broader financial system and become systemic. The major adverse effect has been disruptions to core bond markets. Other interconnections among NBFIs, banks, and core markets can be substantial, too. For example, the rapid expansion of private credit relies heavily on financing and other support from banks. These links undermine one theme of the post-GFC reforms—the transfer of risk away from banks—since, while credit risk has partly migrated to NBFIs, much of the associated liquidity risk remains concentrated in the banking system.

Reflective of its systemic vulnerabilities, NBFI has on several occasions in recent decades, in major jurisdictions, been associated with episodes of financial instability.

These events have primarily involved liquidity spillovers and, in some cases, counterparty failures. Notable examples include the run on US money market funds during the GFC, the global market turmoil of March 2020, and the stress in the UK gilt market in October 2022. While these episodes led—or threatened to lead—to market dysfunction

¹ All monetary amounts are in US dollars.

² NBFIs are defined in this document to include insurance corporations, pension funds, money market funds, hedge funds, and other investment funds; captive financial institutions and money lenders, central clearing counterparties, and other financial auxiliaries (financial market infrastructures); and broker-dealers, finance companies, trust companies, and structured finance vehicles. Classifications vary by jurisdiction and regulatory framework.

and broader financial instability, they were most often contained through large-scale central bank liquidity interventions. Although perspectives differ, there is little doubt that these repeated interventions have reduced agents' perceived risks of funding fragilities. Yet, the increased frequency of such episodes may be a signal of growing difficulties, including in absorbing the large and increasing supply of sovereign debt.

Since the shadow banking collapse during the GFC, NBFI solvency events have been more isolated, and none have been systemic. Nevertheless, cases such as Archegos Capital Management and Greensill Capital, two major financial scandals in 2021 that caused significant losses on counterparty banks, and recent failures, including First Brands and Tricolor, two major companies in the automotive sector, are signs of incipient stress. And the refrain “until the music stops, we keep dancing” being current again suggests systemic vulnerabilities are rising today, warranting close attention.

The risk of systemic events related to NBFI appears to be increasing. A “perfect storm” may be brewing, for several reasons. First, the global financial system already exhibits significant vulnerabilities. Second, past and ongoing policies—particularly highly expansionary fiscal and monetary policies in advanced economies—have encouraged excessive risk taking. Third, economic and financial uncertainties are currently elevated, reflecting shifting geopolitics and increasing fragmentation in trade and finance. Together, these factors make volatility and global financial market stress more likely. Yet stress is more difficult to manage given stretched public balance sheets and limited international policy coordination. The rapid expansion of NBFI and its often-light regulation and supervision is exacerbating this. Moreover, the complexity of NBFI and its opacity—due in part to limited, poor-quality, and unevenly available data—not only hamper timely systemic risk assessment and complicate the identification of critical vulnerabilities but also render stress episodes harder to contain. While views may differ on the urgency, there is little doubt that adverse shocks could trigger a crisis on a scale not seen since the GFC.

More preventive and remedial actions are needed, particularly in some key jurisdictions.

Although NBFI-related fragilities and systemic risks are long recognized,³ policy responses have remained limited. Securities regulators, whose mandates focus primarily on investor protection, may not always assess systemic risks in NBFI as highly as prudential authorities do. Some policy-makers may be overly confident in market resilience. And central banks have often had little choice but to intervene to fulfill their mandates. But the repeated liquidity interventions to support intermediation centered outside the banking sector may have created the perception that such support comes at low cost. Breaking this cycle—and preventing a further buildup of vulnerabilities and systemic risk—will require stronger and more coordinated policy actions.

This report proposes a set of targeted policy measures to reduce the risks of adverse liquidity spillovers and solvency events and to better manage their consequences should they occur.

Recognizing the many benefits of NBFI, the recommendations seek to improve both growth and financial stability outcomes by reducing vulnerabilities and strengthening resilience to stress. A preemptive approach—rather than regulatory easing—can materially lower systemic risk. Key measures to mitigate NBFI-related vulnerabilities that can disrupt core bond markets include limiting leverage and liquidity mismatches and reducing the procyclicality of margining. The report also highlights the risk reduction and resilience benefits of upgrading market infrastructures and related institutions in key jurisdictions.

Effective responses to emerging stress could include improved central bank mechanisms for providing liquidity to selected NBFI activities and entities, thereby reducing the reliance on broad interventions. At the international level, the global financial safety net should be adapted to reflect the larger role of NBFI, including through the maintenance—and potential expansion—of key currency lending facilities and swap lines.

To address solvency spillovers, the report emphasizes maintaining limits on counterparty exposures, enforcing strict supervision of key financial institutions, and expanding stress tests—particularly for interconnections—and

3 Carstens 2021; Knot 2022.

promoting greater burden sharing in recovery and resolution.

The report underscores the need for stronger collaboration among authorities, improved data at both the domestic and international levels, and for monetary and fiscal policies to internalize more risks. It emphasizes that market supervisors should treat financial stability as a core element of their market integrity mandates. Given NBFIs' large and growing cross-border footprint, the report calls for enhanced international cooperation, including jointly conducted stress tests. Noting significant data gaps and limited transparency, the report stresses the importance of increasing the quantity, quality, and availability of NBFI data, including through greater data sharing among authorities within and across jurisdictions. Improved data supports more effective systemic risk analyses and better-targeted interventions and, when accompanied by greater disclosure, also strengthens market discipline. The report highlights the need for monetary policy to better internalize the risk-taking and buildup of vulnerabilities in NBFI. When providing liquidity support, central banks should clearly distinguish between monetary policy and financial stability objectives and communicate transparently how

such interventions are designed, unwound, and terminated. Finally, the report notes that rising NBFI-related risks reinforce the case to build fiscal headroom.

The report is structured as follows. Section I reviews the growth of NBFI, notably of its riskiest forms, and highlights key cross-jurisdictional differences. It also discusses the benefits of NBFI for economic growth and volatility, and analyzes the specific systemic risks associated with NBFI and their underlying drivers. It focuses on liquidity mismatches, high leverage, and the systemic implications of NBFI's two-way connections with core bond markets and banks. Section II examines key structural developments that could amplify NBFI-related risks and assesses current vulnerabilities and signs of potential systemic stresses, particularly for core markets, banks, and across borders. Section III presents policy recommendations to reduce NBFI-related vulnerabilities and limit the risk of adverse spillovers. It also reviews ways to improve the management of liquidity stresses, address the consequences of NBFI failures, strengthen data availability and quality, and the need for monetary and fiscal policies to consider NBFI-related financial stability risks. Section IV concludes with observations on next steps and areas for further analysis.

SECTION I.

Definition of NBFI, and Its Growth, Benefits and Systemic Risks

NBFI, particularly its risky forms, has been rapidly growing in many advanced economies and some emerging markets. NBFI comes with many benefits, but can also raise new systemic financial stability risks, which can be captured in three broad forms.

I.1. Definition, size, and growth

The term NBFI covers a wide range of activities and entities, but there is no universally agreed definition of the term. The Financial Stability Board (FSB), in its annual monitoring report,⁴ defines the sector broadly as all financial institutions that are not central banks, banks, or public financial institutions. It then defines, as a subset, other financial intermediaries as those institutions that are not insurance corporations, pension funds, or financial auxiliaries. Other financial intermediaries include money market funds, hedge funds, and other investment funds, captive financial institutions and money lenders, central clearing counterparties, broker-dealers, finance companies, trust companies, and structured finance vehicles. Investment funds are defined here to cover money market funds and other investment funds, but not hedge funds. Other investment funds include funds active in private credit and private equity.⁵

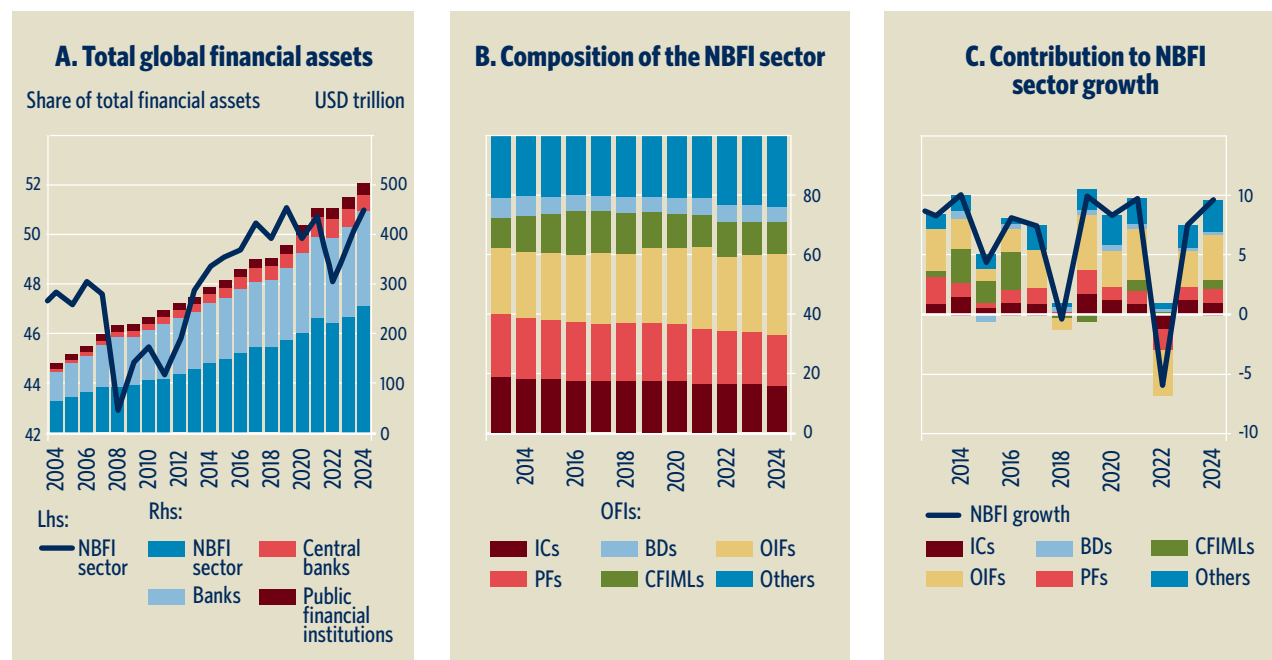
Nonbank financial intermediation has grown rapidly since the global financial crisis. Figure 1, panel A shows the growth of NBFI relative to other forms of intermediation and its overall share during 2004–24. Globally, total nonbank financial intermediation amounted to about \$260 trillion as of end-2024, accounting for more than half of total financial intermediation.⁶ Figure 1, panel B shows the breakdown of nonbank financial intermediation by type. Other financial intermediaries, considered to include the riskier NBFI forms, have globally increased their share of total nonbank financial intermediation by about 5 percentage points over the last decade, with other investment funds recently contributing significantly to the increase in other financial intermediaries, becoming the largest share of other financial intermediaries (41.1 percent). Figure 1, panel C shows the contributions during 2014–24 to NBFI growth by various entity types. It shows that the “other financial intermediary” segment has often contributed much to NBFI’s general growth over the past decade.

⁴ FSB 2025d.

⁵ See further FSB (2013) for definitions.

⁶ As in other parts of the financial system, there is double counting, which may be greater in NBFI.

FIGURE 1. Panel A: NBFIs as a share of total global financial assets increased; Panel B: Composition of the NBFIs sector has shifted; Panel C: OIFs were the largest contributor to the increase of NBFIs assets in 2024



Source: FSB 2025d.

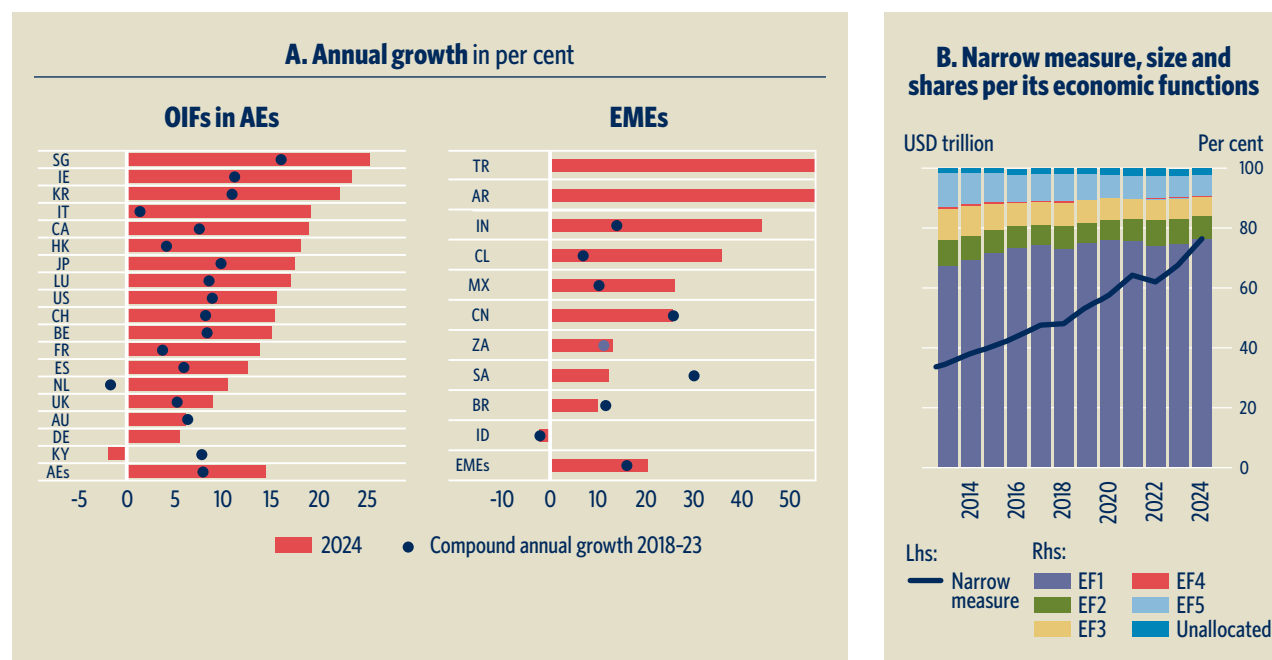
Note: BDs = broker-dealers; CFIMs = captive financial institutions and money lenders; ICs = insurance corporations; LHS = left-hand side; OIFs = other financial intermediaries; PFs = pension funds; RHS = right-hand side.

NBFIs functions with greater financial stability risks have grown especially fast in many jurisdictions.⁷ For 2024, the latest year for which data are available, other financial intermediary's (the riskier NBFIs form) annual growth rates were generally high for both advanced economies and emerging markets, almost all of which exceeded their compound annual growth rates during 2018–23 (figure 2, panel A). NBFIs data can also be partitioned by degree of financial stability risks. The FSB's so-called narrow measure includes entities involved in credit intermediation activities that may pose bank-like financial stability risks (that is, that involve maturity or

liquidity transformation, leverage, or imperfect credit risk transfer) and/or regulatory arbitrage. Figure 2, panel B, shows that this measure doubled between 2014 and 2024, reaching \$76.3 trillion. Figure 2, panel B also shows that about three-quarters of the narrow measure represents today what the FSB calls economic function 1 (EF1), that is, collective investment vehicles with features that make them susceptible to runs (for example, money market funds, fixed income funds, exchange-traded funds, and credit hedge funds). This function has driven much of the narrow measure's recent growth via net inflows and higher valuations.

⁷ Because regulatory and reporting frameworks vary widely across jurisdictions, comparability of measures of NBFIs has its limits. Furthermore, while classification by entity is most used, it has drawbacks. It may miss commonalities across types and overlook diverse entities engaging in similar activities. It is also inherently static, taking current forms as given, yet entity boundaries change continuously due to competitive, innovative, and regulatory forces. Complementary perspectives come from classifying by activity (for example, credit extension, deposit taking, securitization, use of derivatives) or economic function (for example, risk management, maturity transformation, price discovery). Classifying by activity (or function) allows examination of developments, including risks, without the need for major changes in the analytical toolkit, and can provide a more holistic view. Entity and activity types can overlap, such as when one entity exclusively provides one activity (for example, leasing companies), but functionally equivalent services can be provided by distinct entities (for example, both open-ended money market funds and banks provide deposit-like instruments). Data collection and analysis by activity and functions are more difficult (see Cetorelli et al. [2025] for an effort to classify NBFIs by activity).

FIGURE 2. Panel A: Other Investment Fund (OIF) size and growth by jurisdiction; Panel B: Size and shares of the narrow measure, per economic function.



Source: FSB 2025d.

Note: AEs = advanced economies; EMEs = emerging markets; EF = economic function; LHS = left-hand side; RHS = right-hand side.

I.2. Benefits and systemic risks

There are many benefits and risks associated with NBFI, as conceptual and empirical evidence shows, mostly similar to other forms of financial intermediation. Collectively, however, NBFI can raise systemic risks, with specific causes, channels, and manifestations.

NBFI has many benefits for economic growth and societies. In principle, the growth of NBFI reflects, at least in part, the demands of various economic agents—households, firms, and sovereigns—for its type of intermediation. Indeed, NBFI has greatly expanded its financing of both the public and private sectors in many jurisdictions. In advanced economies, it has become a major funding source for governments, as commercial and central banks have retreated. Its growing role in financing the private sector is exemplified by private credit, which

in some jurisdictions is now surpassing traditional bank financing to nonfinancial corporations. Besides more financing for investment and consumption, associated possible gains include better-matched forms of finance and possibly better resource allocation, leading to higher economic growth. This view also means that NBFI's benefits relative to those of banks may grow as countries develop. Banks are well suited to provide relationship-based forms of financing and liquidity services, which are important for small firms and in environments with greater information asymmetries. But, as countries develop, the contribution of banks declines and that of NBFI grows.⁸

More NBFI could also help lower real and financial volatility, as it tends to absorb shocks better. NBFI generally uses instruments that share asset-related risks (including insolvency risks) among a wide pool of investors. And as many NBFI entities pool

⁸ Growth opportunities and sources of new productivity could reside more in sectors that NBFI is better at financing and promoting (for example, intangibles-intensive activities). NBFI could also represent more diverse views and trading motives than those of banks, potentially making for better price discovery. See further Claessens (2016).

idiosyncratic risks at the intermediary level, they only pass residual risks on to their final investors. Furthermore, many NBFIs do not have levered balance sheets and are not core to the payment system. This makes individual NBFI failures less likely to have systemic consequences than bank failures typically do. Indeed, conceptual and empirical evidence shows that by acting as a “spare tire,” countries with more diverse financial systems, including more NBFI, often recover faster from banking crises and asset price slumps.

NBFI’s benefits, though likely, vary across its many types and are less proven. To assess risks properly and allocate resources efficiently, gains must be seen relative to the ability and incentives of individual entities. Some argue that credit risk analysis and due diligence for some NBFI types leave much to be desired. To the extent that it is worse than banks, gains from better resource allocation through more NBFI are elusive. Also, when a specific type only passively invests in corporate securities (for example, as in money market funds), its contributions to growth may be limited to providing financing, as this adds little to improving firm-specific governance or information. Lacking access to central bank liquidity, NBFI can also add less to overall liquidity than banks can. To date, many of these gains have not been analyzed in detail. Furthermore, with all its heterogeneity, it is difficult to definitively state NBFI’s aggregate benefits.

Collectively, NBFI comes with distinct financial stability risks. NBFI’s systemic risks largely arise from its major liquidity mismatches, notably of the various investment fund vehicles, and the high leverage of some types. The underlying and proximate causes of such vulnerabilities are well known and are largely the same as those for banks. They typically include poor incentives,

poor regulation and weak supervision, tax and legal distortions, and possible adverse side effects from monetary and fiscal policies. As for banks, but more likely for NBFI, given its lighter regulation and oversight, these factors can mean weaker credit and liquidity risk management. Most important, however, is that the externalities and market failures that can lead to system-wide excessive liquidity mismatches and high leverage are greater, but less internalized for NBFI than for banks, given its above-mentioned lighter and less financial stability-oriented regulation and supervision. In the face of such vulnerabilities, relatively small shocks can easily trigger stress and lead to systemic NBFI-related events.

What differs from banks is that the systemic consequences of NBFI mostly arise from its connections with core bond markets and banks.

The main financial stability concerns arise from NBFI’s adverse spillovers on core bond markets and banks.⁹ Liquidity mismatches imply risks of runs, as in the case of investment funds, here defined as money market funds and other investment funds; and shocks can trigger large margin and collateral calls, as in the case of hedge funds and pension funds. Liquidity shortfalls, which can be triggered by many factors, with leverage acting as an amplifier, can then adversely affect core bond markets in systemic ways. Leverage in NBFIs is generally less than for banks, and insolvency-related losses are, in contrast to banks, typically shared, immediately or over time, with NBFI investors, beneficiaries, and other stakeholders, including sometimes fiscal authorities. Still, for some activities and entities, leverage can be high, and when insolvency-related losses are not absorbed smoothly by NBFI stakeholders, they can trigger systemic spillovers. Table 1 summarizes these risks.

9 FSB 2022b, 2025d; BCBS 2025. BCBS (2025) analyzes four specific scenarios: 1. Stress among NBFIs leads to credit losses or liquidity pressure on banks and other market participants, as in the case of credit losses due to Archegos or the liquidity stress from investment fund redemptions in March 2020 and the UK pension fund margin calls in October 2022; 2. The failure of an NBFI impacts the parent banking group, as happened in some emerging markets with nonbank financial corporations; 3. NBFIs stop taking risks from banks that depend on them, as happened during the GFC due to the failure of insurance company AIG; and 4. NBFIs stop providing funding to banks, as money market funds did in March 2020’s “dash-for-cash.”

TABLE 1: Conceptual framing and examples of systemic risk related to NBFI

Manifestations	SOURCES OF SYSTEMIC RISK		
	Within NBFI		Spilling over to the main system through interconnections
	Leverage (defaults)	Liquidity (shortfalls)	
Channels of systemic impact	Large NBFI defaults, leading to credit losses at banks.	- IFs' liquidity needs. - HF's PFs, ICs, and other NBFIs' margin and collateral calls.	- Losses at banks and public finance impact. - Market dysfunction.
Past examples	- Shadow banking. - Greensill, Archegos, First Brand, other NFCs, some NBFCs.	- IFs (GFC, March 2020). - PFs (UK LDI crisis). - HFs (various). - Margins (March 2020, March 2022)	- Shadow banking (GFC). - Various core market dysfunctions (GFC, March 2020, March 2022).
Potential future examples	- Losses of ICs, PFs, and NBFCs. - Large NFC losses.	- Large redemptions and liquidity shortfalls (IFs). - Margins, collateral calls (HFs, PFs, ICs).	- Private credit, other large defaults leading to credit losses at banks. - Liquidity needs of IFs, HFs, ICs, and PFs.

Note: HFs = hedge funds; ICs = insurance corporations; IFs = investment funds; NBFCs = nonbank financial corporations; NFCs = nonfinancial corporations; PFs = pension funds.

Liquidity-related stresses related to funds have become more common. Events include the outflows from money market funds during the GFC and repeated stresses in core bond markets, including in March 2020 (the “dash-for-cash”), March 2022 (Russia’s invasion of Ukraine), and, to a lesser extent, events in March 2023 and April 2025. Most adverse liquidity spillovers have originated from funds. When faced with large-scale redemptions and other withdrawals, funds can quickly run down their buffers. At such times, fund managers typically also hoard cash. Funds may then demand liquidity services from banks, but banks’ supply is not fully elastic (as they do not have the balance sheet capacity or willingness to lend). Faced with these liquidity shortages, funds have to sell assets at times with few buyers. Depending on the size and concentration of investments sold, this can lead to fire sales and market imbalances with dysfunctions, with spillovers to other parts of the financial system and the real economy. Liquidity stresses can also relate to rapid capital movements by internationally active NBFIs, as has occurred in emerging markets (for example, the Republic of Korea, Mexico, and Türkiye).

NBFI’s hedging and derivatives strategies can also lead to large liquidity needs during periods of market volatility. Notably, hedge funds’ various relative value strategies are highly vulnerable to adverse shocks in funding, cash, or derivatives markets. Shocks can also relate to spikes in margin calls¹⁰ and can get amplified by the high leverage of NBFI entities. Past episodes evidence these risks. Triggered by a shortfall in central bank reserves, on September 17, 2019, hedge funds’ large-scale basis trade arbitrage strategy faced immense pressure, and repo rates spiked to 10 percent. This created significant margin stresses as cash became scarce and other funding options were limited. During the market turmoil of March 2020, margin calls in Treasury futures markets triggered an unwinding of the basis trade, including of holdings of Treasuries in the cash market, contributing to destabilizing deleveraging spirals. Large margin calls worsened price movements and led to adverse liquidity spillovers and stresses in commodity markets in March 2022 when energy and other prices spiked following the invasion of Ukraine. In the liability-driven investment crisis in the UK gilt market¹¹ in September 2022,

¹⁰ BCBS-CPMI-IOSCO 2022; 2025.

¹¹ The gilt market refers to the debt securities issued by the Bank of England on behalf of His Majesty’s Treasury, whose paper certificates had a gilt, or gilded, edge.

as interest rates rose following a budget announcement, pension funds faced large margin calls on their derivatives, forcing sales of UK gilts, which, in turn, led to market disruptions, which required central bank support. More recently, in early April 2025, the unwinding of relative value trades—tied to interest rate swap markets, where investors had bet on a narrowing in spreads—contributed to the heightened volatility observed in the US Treasury markets.

As events have led, or threatened to lead, to systemic consequences, central banks have on several occasions provided various forms of liquidity and other support.

In most cases, they bought large amounts of sovereign and other bonds to inject liquidity and support asset prices (for example, the September 17, 2019, event prompted the US Fed to inject \$75 billion in liquidity into the repo markets that day and every day for the rest of the week). In other cases, they engaged in large or even open-ended (March 2020) purchases of sovereign and other bonds to inject liquidity. In some cases, they established or expanded (repo) facilities specifically designed for NBFIs, and in other cases, central banks provided foreign currency liquidity during market disruptions related to nonbank financial intermediation (for example, during the early stages of the Covid-19 crisis in Korea, through repurchase agreements).

While leverage is often an amplifier, pure insolvency-related systemic events have been fewer.

Leverage in NBFI often amplifies adverse shocks¹² but is less likely to lead to large losses for the core banking system through connections. In a few cases, large NBFI insolvencies threatened adverse effects on core banks, but no public support was needed. Examples include Archegos and Greensill in March 2021¹³ and, recently, for the US, First Brands (car parts) and Tricolor (subprime car lender). Also, emerging markets have seen defaults of (bank-affiliated) nonbank financial corporations, with related

spillovers.¹⁴ Only in the GFC did many banks, directly or indirectly, need support because of large scale failures in the shadow banking system, then, a major part of nonbank financial intermediation.

1.3. Three broad NBFI forms clarify specific drivers and related types of systemic risks

Since NBFI straddles many institutions and activities, each with specific benefits and risks, it is preferable to focus on broader forms rather than on detailed types, which can change rapidly. This allows for a clearer consideration of what drives NBFI growth, each broader form's uniqueness, and its potential systemic risks. Such framing informs, among others, whether the growth of a form reflects genuine demand for its services or excessive risk-taking and regulatory arbitrage by the suppliers.

Three broad forms considered by Acharya, Cetorelli, and Tuckman (2026a; 2026b) are parallel, substitute, and transformation, as summarized in table 2.¹⁵ The *parallel* form captures those distinct activities that banks do not offer and that operate largely separately from banks and do not compete directly with them. Many traditional activities of insurance corporations and pension funds are in this form (for example, [super] annuities). The diametrically opposite *substitute* form does what banks traditionally have done but now competes with or migrates activity from them. Many types of money market funds are in this form, such as those that largely invest in government securities, as they, like bank deposits, are safe and highly liquid.

The *transformation* form has grown considerably in recent years. It provides new types of services in part by complementing bank-based services, but also by transforming them. This form captures fast-growing private credit. In many dimensions, private credit is similar to bank loans. For example, it also displays low liquidity and

12 FSB 2023b; 2025a.

13 Archegos Capital Management was a private hedge fund that collapsed due to over-leveraged bets, while Greensill Capital specialized in high-risk supply chain finance, both exposing significant risk management failures. The collapses of Archegos and Greensill led to large losses at Credit Suisse, contributing to its subsequent crisis and demise, and at other banks.

14 See, further, BCBS (2025).

15 Some types of NBFI activities and entities could take multiple forms. The mix also varies by country. In some jurisdictions, banks are more limited in what they can do, but in others, banks can engage directly in a wide range of activities, including what elsewhere could be classified as NBFI, or own various NBFI entities (for example, through holding companies). Key is that the classification is dynamic. Traditionally, for example, banks originated syndicated and other large loans and then sold these (largely) to NBFIs. But now, NBFI originates and extends such loans directly.

often stale valuations. However, private credit does not fully displace bank lending but rather transforms it. That is, while NBFIs lend to nonfinancial corporations, banks still provide these entities with credit and liquidity lines and other services, creating various links and mutual dependencies. The transformation form also includes proprietary trading, such as in the basis trade. This activity has, in part due to regulatory changes, moved from banks to

entities like hedge funds, but it still relies on bank funding through repos. Transformation implies that risks cannot be understood by examining banks and NBFIs separately; while credit risk may migrate (in part) from banks to NBFIs, liquidity risk remains concentrated in banks. And, as the interconnectedness between the two sectors grows, the risk of (implicitly) expanding the public safety net that supports banks, and core markets, increases.¹⁶

TABLE 2: Three broad NBFI forms: examples, main drivers, and sources of systemic risk

Form	Parallel	Substitute	Transformation
<i>Examples</i>	PF and IC traditional products (e.g., pensions, annuities).	IFs (e.g., MMFs investing mainly in sovereign securities).	Private credit, shadow banking. HF's basis trade.
<i>Likely most important driver(s)</i>	Overall economic and financial conditions.	Technology, financial conditions, and regulations.	Regulations and supervisory practices, technology.
<i>Main type of systemic risk</i>	Leverage (underfunding) to banks, public finance. Sometimes liquidity.	Liquidity, through interconnections to core bond markets.	Interconnections, notably, to banks, via leverage and liquidity.

Source: Based on Acharya, Cetorelli, and Tuckman (2026a; 2026b).

Note: HFs = hedge funds; IC = insurance corporation; IFs = investment funds; MMFs = money market funds; PF = pension fund.

Importantly, the drivers of growth differ by broad forms. Drivers generally include (changes in) regulation and supervision, technology, monetary and other financial conditions, and economic growth and related demand for specific forms of financing. But each form is affected by a different mix (see table 2). Tighter bank regulation and supervision affect the parallel form little, while regulatory arbitrage matters most for the transformation form (for example, studies suggest that private credit growth reflects regulatory arbitrage, similar to what drove shadow banking before the GFC.¹⁷) Technological advances matter for investment funds—by allowing for wider and cheaper distribution—and to some extent for

private credit—by making processing information easier, but less so for other activities.

Crucially, each form comes with specific systemic risks. Distinguishing broad forms, rather than using the narrower NBFI types, and investigating their specific drivers, can help identify potential vulnerabilities and how they can lead to systemic risks. The main systemic risk for the parallel form tends to arise from high leverage, for the substitute form from liquidity imbalances, and for the transformation form through interconnectedness.

16 Relatedly, while NBFIs' institutions and activities are heterogeneous, strategic complementarities can increase commonalities in behaviors, including in liquidity mismatches and market exposures, heightening risks of fire sales, as well as adverse spillovers (Cetorelli, Landoni, and Lu 2023) and bailouts (Farhi and Tirole 2012).

17 Avalos, Doerr, and Pinter (2025) find credit growth to be bolstered by better funding competitiveness due to differential regulatory treatments, including the relative treatment of banks' on-balance-sheet compared to off-balance-sheet exposures, while Chernenko, Ialenti, and Scharfstein (2025) find the capital treatment of loans to NBFI private credit providers to be relatively favorable, enabling banks to exploit their low-cost funding.

SECTION II.

Key Developments Combine with NBFIs Growth to Increase Systemic Risks

The risk of systemic events related to nonbank financial intermediaries (NBFIs) is increasing as several structural developments combine with the rapidly growing vulnerabilities of NBFIs. A “perfect storm” may be brewing.

II.1. Key structural developments raising NBFIs-related systemic risks

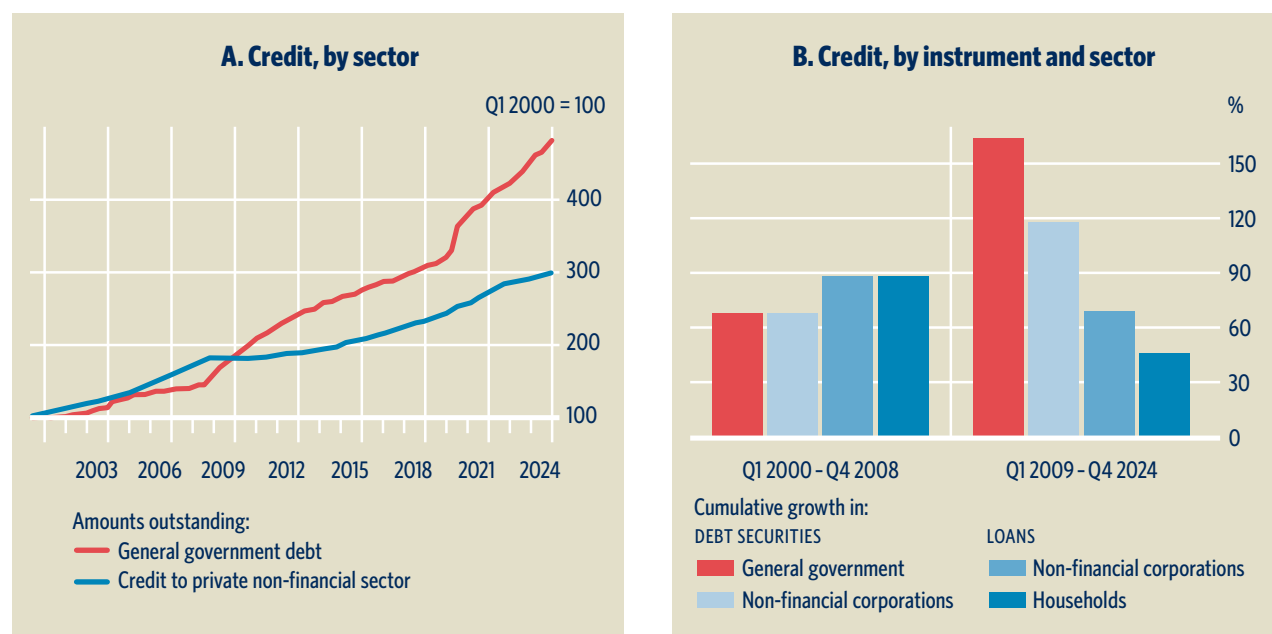
Several structural developments increase the risks of systemic events, including the large increase in sovereign bonds of many advanced economies; the increasing role of NBFIs investors in sovereign financing; the growing importance of NBFIs in private sector lending; the increasing international dimensions of NBFIs; and the large links of NBFIs with banks.¹⁸

Government bond issuance has expanded significantly. Large and persistent post-GFC deficits in

many advanced economies and some emerging markets, further boosted during the Covid-19 pandemic, meant that sovereign debt expanded over the last 15 years some 60 percent faster than credit to the private sector (figure 3, panel A). And whereas in the eight years before the GFC, debt securities and loans to nonfinancial corporations and households grew faster than sovereign debt, in the 15 years since, this has reversed (figure 3, panel B). For a decade, unconventional monetary policies—central bank commitments to keep policy rates low and large-scale bond purchases (quantitative easing)—accommodated this in advanced economies. While only some emerging market central banks and some small open economies bought their own sovereign bonds, many boosted their foreign exchange (FX) reserves by purchasing the bonds of core countries. Combined, the increased holdings of central banks accounted for 63 percent of the increase in G7 sovereign debt between 2008 and 2021.

¹⁸ This section draws on BIS (2025a), Hernández de Cos (2025), FSB (2025d), and IMF (2025a), notably for various figures.

FIGURE 3. Government bond issuance has outpaced lending to the private sector



Source: Hernández de Cos 2025

As central banks tightened policy in recent years, sovereign financing has shifted to NBFIs.

Between 2021 and 2022, to combat high inflation, central banks worldwide sharply raised their interest rates. More recently, major central banks tapered their sovereign bond purchases and let maturing bonds run off, with some selling bonds. With FX reserves stable overall, foreign official demand for core sovereign bonds was also lower.¹⁹ Since the GFC, bank-sovereign links have been discouraged by market discipline, regulatory responses (notably a lower leverage ratio, with supplemental restrictions for systemic banks), and tighter oversight, restraining many banks, too, from buying more sovereign debt. This lower demand has left more sovereign bonds for NBFIs to buy, increasing their share of advanced economy sovereign bond holdings by about 7.9 percentage points between 2021 and 2024 (figure 4). These NBFI investors include short-term-oriented investors, such as investment funds, exchange-traded funds, and hedge funds; and longer-term, more buy-and-hold investors, such as pension funds,

insurance corporations, and sovereign wealth funds. With fiscal deficits and rates higher in major advanced economies than in earlier decades, (re)financing needs are large and NBFIs must remain major buyers.

NBFI financing of the private sector has also been increasing.

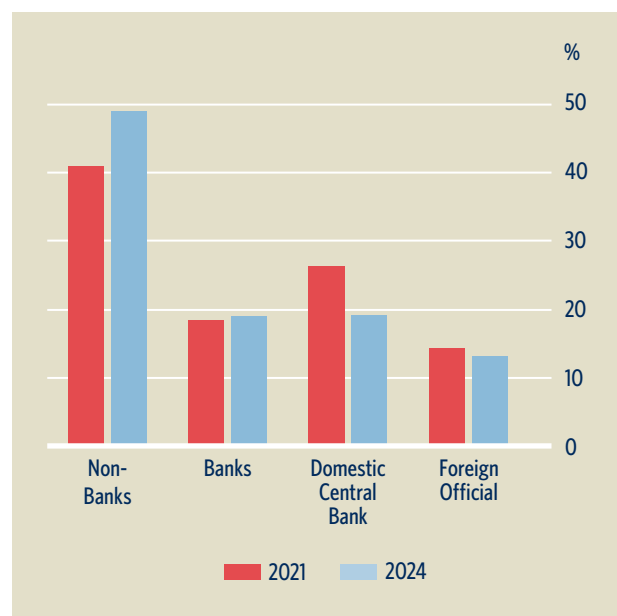
While following post-GFC reforms, some NBFI types, like money market funds, have grown more slowly, other forms, some risky, have been rapidly increasing. Notable is the growing role of NBFIs in extending private credit. While existing for some time in the US, Europe is undergoing a similar trend, with banks pulling back and private lenders, including some from abroad, stepping in to fill the gap. The total size of private credit assets under management, which was estimated to be \$2.1 trillion in 2024,²⁰ amounted, at end-2025, to about \$3 trillion to \$3.5 trillion.²¹ These shifts reflect, in part, tighter bank regulation and supervision since the GFC, but also technological advances that favor more

¹⁹ Eren, Schimpf, and Xia 2025.

²⁰ IMF (2024a), using a narrow definition.

²¹ According to industry surveys, such as ACC (2025).

FIGURE 4. Holders of advanced economy government debt (% of holdings)



Source: Arslanalp and Tsuda 2014, updated.

arms-length financing (for example, easier credit scoring and originate-to-distribute models²²).

Patterns of cross-border capital flows and ownership have become more NBFi-driven.

Pre-GFC, cross-border intermediation occurred primarily through internationally active banks and in the form of loans and deposits, directly to and from corporations and sovereigns. Recently, as NBFIs operate internationally, a considerable amount of cross-border intermediation occurred through NBFIs actively buying and selling sovereign bonds and other debt instruments of both advanced economies and emerging markets.²³ Capital surplus

countries with less vibrant capital markets invested a considerable amount over the last decade in markets overseas (notably the US), as exemplified by the large (net) changes in bond holdings of Europe (figure 5). And internationally active NBFIs have become the marginal buyer for many sovereigns. Much of NBFIs' intermediation also occurs using collateral-based lending and borrowing,²⁴ and is supported by derivatives.²⁵ More generally, nonbank financial intermediation heavily uses global capital markets, creating many direct and indirect lending, ownership, and other links, making for complex connections among countries with limited (regulatory) visibility.

Although large banks remain global players, their roles have evolved.

In the past, banks were key in on-balance-sheet intermediating, albeit not always effectively. Today, banks increasingly transact in various (traded) assets and claims and manage related exposures and risks. This includes facilitating, directly and indirectly, the growing footprint of nationally and internationally active NBFIs through various two-way links. On-balance-sheet lending to NBFIs has notably grown through financing NBFi vehicles engaged in private credit extension and securities financing to hedge funds. Off-balance-sheet exposures have increased through contingent liquidity and credit lines, FX swaps, and new instruments (recently, synthetic risk transfers). Banks also increasingly provide nonbank financial intermediation with clearing, market-making, underwriting, and derivatives services. And NBFi is a major source of (wholesale) funding for banks. If, as is often believed, risk analysis and due diligence are weaker in NBFi, these links can negate the gains of moving risky forms of intermediation out of banks. And they add to complexity and worsen opacity.

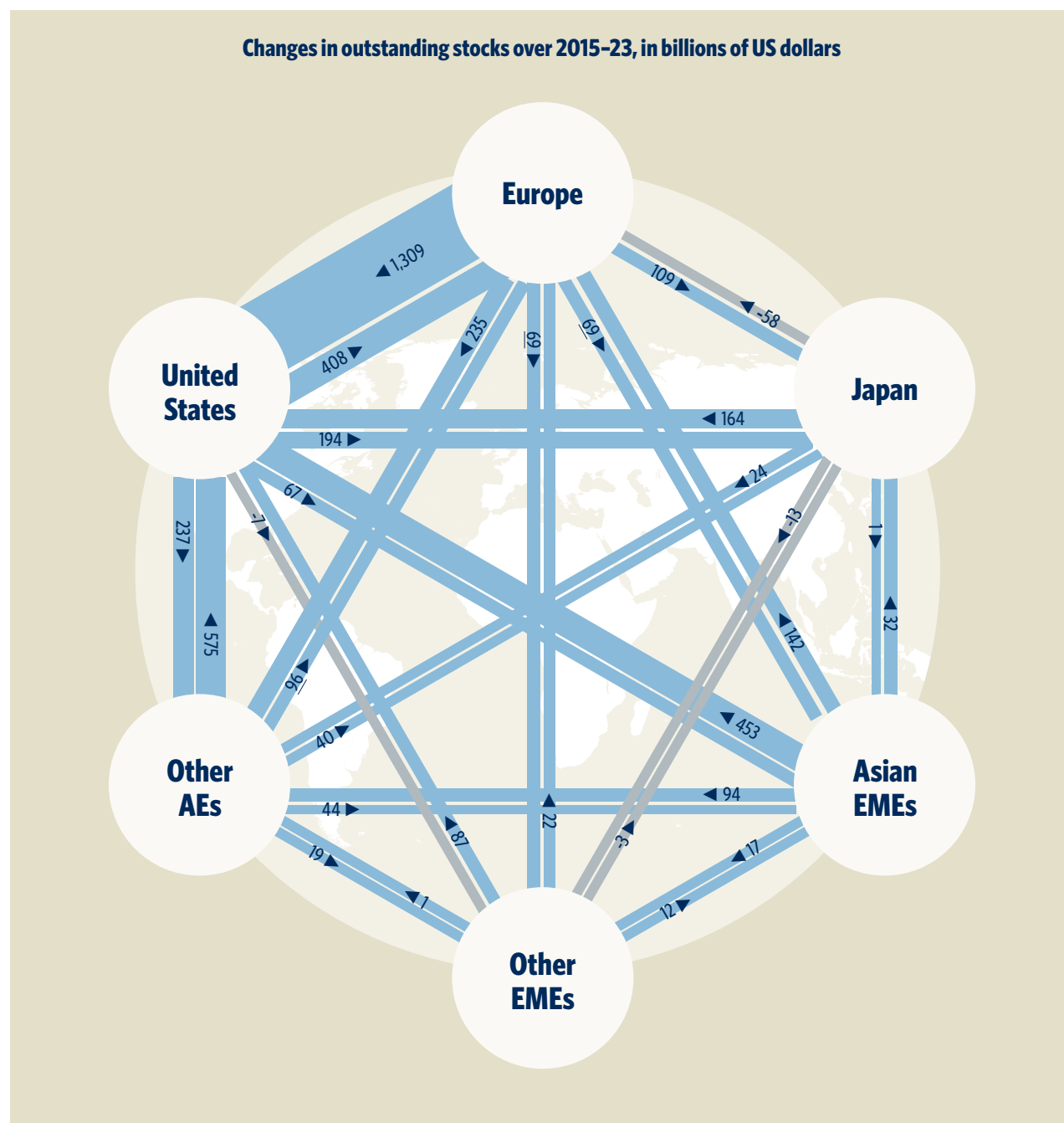
22 The originate-to-distribute model is a strategy where financial institutions originate loans—such as mortgages or corporate debt—and quickly sell them to investors via securitization or syndication, rather than holding them until maturity. This shifts risk off their balance sheet, increases liquidity, and allows faster, volume-driven lending, but can weaken underwriting standards and create moral hazard.

23 For example, Chari 2023; Buch and Goldberg 2024; Pelizzon et al. 2025; see, further, BIS 2025a.

24 Singh 2020.

25 BIS 2025a.

FIGURE 5. Changes in global cross-border bond holdings, excluding official reserves



Source: BIS 2025a.

Note: AEs = advanced economies; EMEs = emerging markets.

II.2. The many signs of growing NBFi-related vulnerabilities

NBFI stress is episodic, since lacking natural backstops, liquidity can evaporate quickly. And stress can be triggered by many shocks, from idiosyncratic events (for example, a failed auction) to some policy or other announcements. Stress periods are also often short-lived, in large part because of the (to date) rapid central bank interventions. Relatedly, market-based indicators do not change significantly ahead of NBFI-related stress episodes, and few other indicators reliably signal risks sufficiently far ahead—certainly much less so than for bank stresses.²⁶ Nevertheless, and focusing on the most important risks, liquidity, and leverage shortfalls, and related spillovers, several indicators suggest growing vulnerabilities, and several recent events reveal signs of incipient stress.

Vulnerabilities in core markets are high and increasing. Sovereign bond markets are the backbone of financial systems. Stress in markets with many links can especially lead to sharp asset price changes and spikes in risk sentiment globally. Their resilience is thus of global importance. But the structures of core bond and other funding markets have changed dramatically since the GFC. In part due to regulatory changes, the liquidity-supply ecosystem has shifted away from traditional dealers and increasingly relies on intermediaries with fragile funding structures and opportunistic investment strategies. Internationally active hedge funds, high-frequency traders, and asset managers are big players, trading and taking large, short-term

positions (for example, the basis trade). Relatedly, the repo markets have become very large (for example, \$13 trillion daily in the US), and, while it slightly retrenched after the stress of March 2020, prime brokerage and repo borrowings in the US have doubled over the past two years.²⁷ Hedge funds operate in this market with high leverage, up to 200 (as haircuts can be just 50 [or even zero] basis points) (figure 6).²⁸ With funding, often intermediated by banks and at very short-term maturities, hedge funds are prone to shocks, leading at times to sudden retrenchments, amplifying imbalances and causing procyclical deleveraging and market dysfunctions.²⁹

Systemwide, NBFI links with banks are growing.

Data show growing connections between NBFIs and banks, to the extent they can be documented. Using publicly available data, the IMF³⁰ reports that many US and some euro area banks have on-balance-sheet exposure to NBFI that much exceed their capital (figure 7, panel A). Many other links, especially those off balance sheet, are harder to document and may be underestimated, but data show high exposure, too (figure 7, panel B). In Europe and the US together, loans to all NBFIs were, at end-2024, on average, 9 percent of bank loan portfolios, and total exposures were about \$4.5 trillion, of which \$2.6 trillion were loans and the rest undrawn commitments.³¹ As noted, these links are not side effects of the growth of NBFI but are, rather, part of their transformation, and confirm the risks of possible adverse spillovers from NBFIs to banks.

26 For banking stress, which is both more frequent and more studied, various early warning indicators exist. Still, predicting banking crises is difficult, with large Type I and II errors. (Type I and Type II errors refer to prediction mistakes. A Type I error occurs when a crisis occurs but was wrongly not predicted. A Type II error happens when a crisis was predicted but wrongly does not occur.) Aldasoro, Hoerdahl, and Zhu (2022) develop market condition indexes but find them to have limited predictive (and explanatory) power for (future) market stresses, although somewhat better when using machine learning (Aldasoro et al. 2025). And even ex post, it is often challenging to explain NBFI-related systemic events.

27 Data from the Office of Financial Research, US Department of the Treasury. See, also, Cenicola et al. (2025) and Sushko and Todorov (2025). Data collected from public reports also show the large increase in overall pledged collateral received by major banks for reuse (see Singh 2020; 2026a).

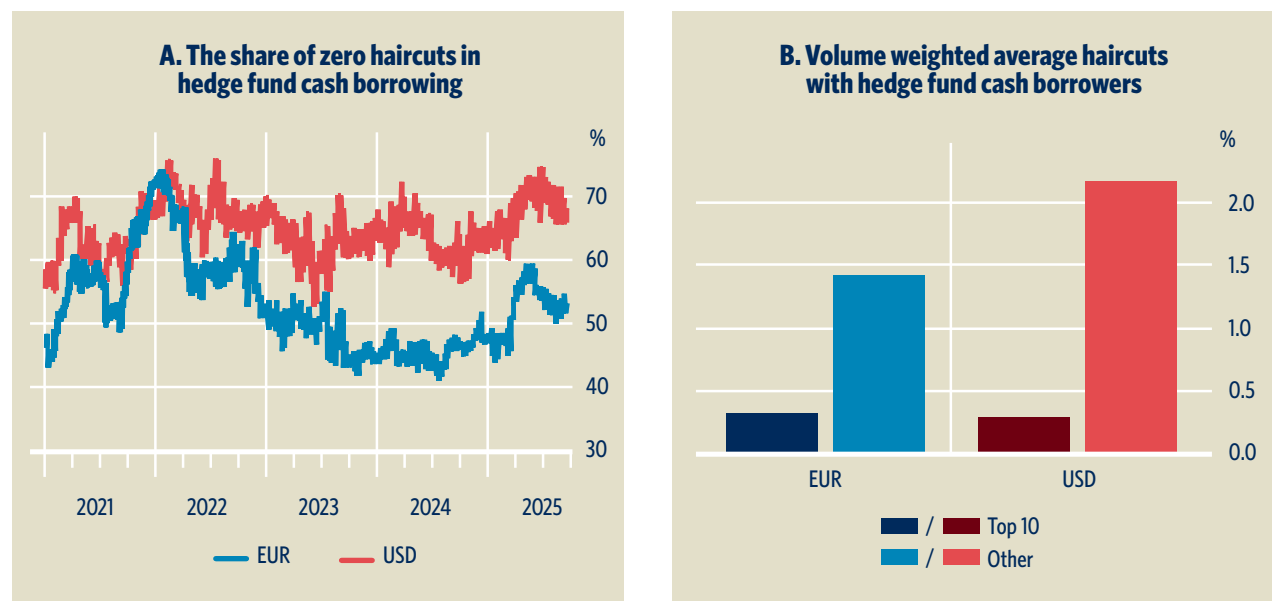
28 Creditors can set bilateral haircuts low, even negative, for lending to hedge funds, as they have other assets and business with the hedge funds (see Bejerano, Lu, and Wallen 2026).

29 See, further, Adrian, Fleming, and Nikolau (2025) and FSB (2026a).

30 IMF 2025a.

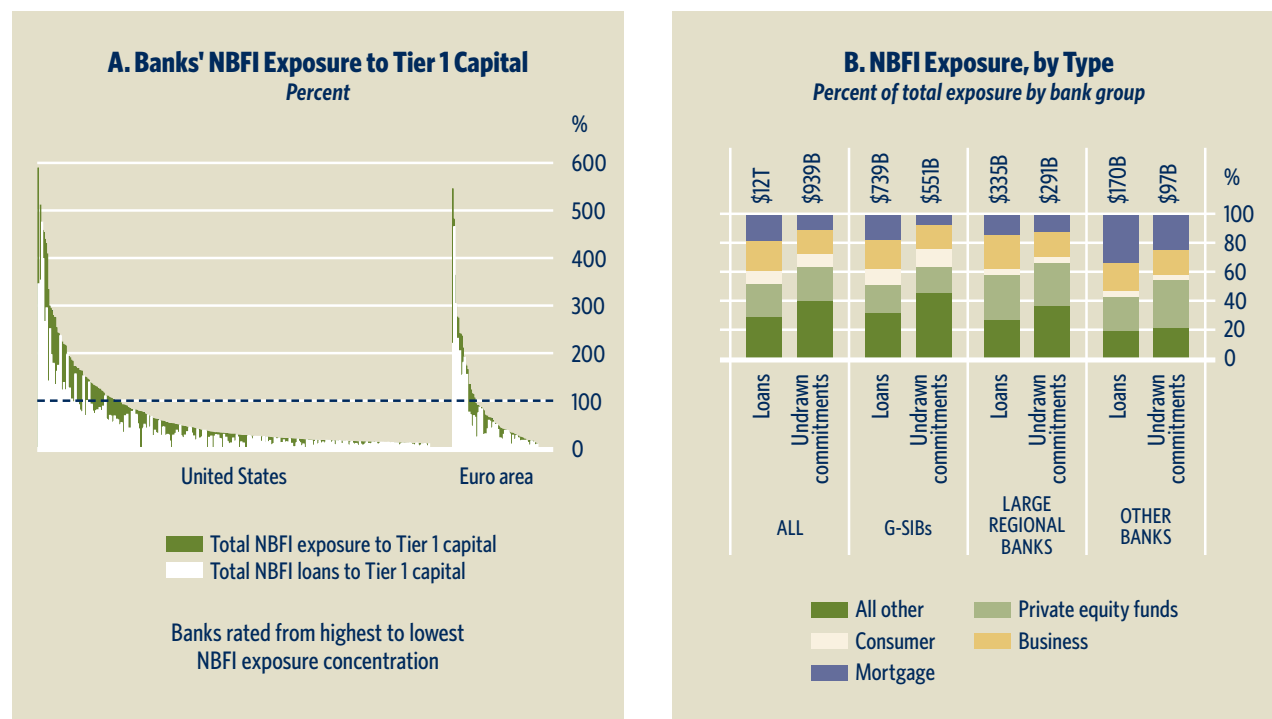
31 IMF 2025a. For the US and covering private credit only, Berg and Lee (2026) estimate that at end-2024 total bank and nonbank lending to private credit entities, including business development companies, ranged between \$410 and 540 billion and capital call obligations were \$300 billion. For bank-NBFI links, see further also FSB (2025d).

FIGURE 6. Low haircuts are concentrated among large hedge funds



Source: Hernández de Cos 2025.

FIGURE 7. Bank Exposure to Nonbank Financial Intermediaries



Source: IMF 2025a.

Other NBFi risks and intra-NBFi links are growing, as well.

Insurance corporations (ICs) and pension funds have grown in size and have broadened their investments in recent years, adding riskier and less liquid, often not publicly rated, assets, including private credit, and use more derivatives, often with synthetic leverage.³² ICs increasingly use nontraditional liabilities and engage more in reinsurance.³³ Intra-NBFi, ICs as well as pension funds, are increasing their links with other NBFi types.³⁴ ICs have developed ownership and control links with private equity and credit, raising potential conflicts of interest.³⁵ Other connections include forms of contingent support and guarantees. And there are signs of regulatory arbitrage through intra-NBFi links.³⁶ Price data suggest greater interconnections of ICs with banks.³⁷ All this increases systemic risks and heightens opacity.

Several related events suggest rising systemic risks.

Besides the recent defaults of First Brands and Tricolor,³⁸ there are indications of incipient stresses in the private credit and equity markets. Several business development companies that had raised public equity from retail investors recently similarly faced withdrawals, in part due to scrutiny about the opacity of underlying assets and related valuations, and some have been forced to liquidate positions. The large redemptions that some funds (for example, Blue Owl, Blackrock, BlackRock, and Cliffwater) recently have experienced led their management to restrict withdrawals and some owners to inject new equity. And many

business development companies now trade at significant discounts from their net asset values. That even entities with relatively low leverage are experiencing redemption risks suggests possible fire sales and contagion to other similar firms and rising systemic risk.

Reflecting its growing global footprint, the contribution of NBFi to cross-border spillovers is increasing.

Besides through NBFi-related capital flows, NBFis are now more active in global funding markets and more interconnected internationally than traditional intermediaries are, with currency forwards,³⁹ swaps, and other derivatives key instruments facilitating linkages. While activities can reduce risks, for example, when hedging lowers FX risk, they have short maturities, exposing participants to liquidity risks. One stress amplification channel stems from the close links between the repo and FX swap markets.⁴⁰ Another channel relates to actions of long-term private investors, such as asset managers, pension funds, and insurance corporations. The unwinding of trades can strengthen transmissions, particularly when carry trade strategies⁴¹ are used and risk appetite recedes.⁴² Whereas major markets' spillovers used to impact just some economies, today they impact many jurisdictions, making core asset prices and other financial conditions closely linked globally (figure 8). With regulatory and reporting frameworks and public support mechanisms increasingly varying across jurisdictions, the scope for stress in one core market to spread in unexpected ways globally risks becoming larger.

32 IAIS 2024, 2025; Aquilina et al. 2025b.

33 Garavito et al. 2025.

34 FSB 2025e.

35 Cortes et al. 2025; IAIS 2025; IMF 2025b.

36 Insurance corporations increasingly move risky assets to off-balance-sheet entities and pursue so called asset-intensive reinsurance (often with hedge funds and pension funds and largely through offshore entities) for regulatory relief while still retaining exposures, adding to complexities.

37 For example, Aquilina et al. (2025b).

38 First Brands Group, a global automotive parts manufacturer, and Tricolor Auto Group, a subprime auto lender, were two major companies in the automotive sector that filed late 2025 for bankruptcy.

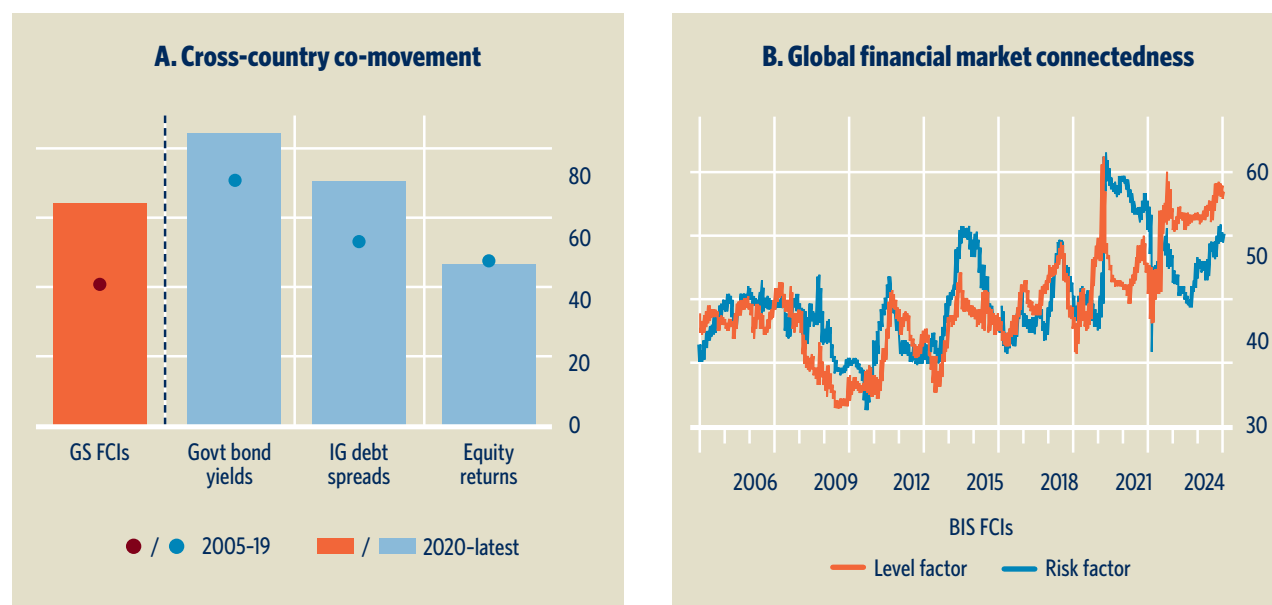
39 A currency forward (or FX forward) is a binding, customizable over-the-counter agreement between two parties to exchange a specific amount of one currency for another at a fixed rate on a set future date. They are primarily used for hedging.

40 While not highly leveraged, when using short-term FX swaps to hedge currency risk for long-dated securities, intermediaries face considerable short-term funding rollover risks (Nenova et al. 2025).

41 A carry trade strategy involves borrowing in a low-interest-rate currency and investing in a high-interest-rate asset, trying to profit from the interest rate differential (the "carry").

42 Following the April 2, 2025, US tariff announcement, moves in the dollar exchange rate were exacerbated (Shin, Wooldridge, and Xia 2025). The unwinding of yen carry trades in August 2024 was another example (BIS 2025a; Packer et al. 2024). While both turbulences were relatively short-lived and had limited effects, they illustrate how vulnerable carry trades are to rapid reversals.

FIGURE 8: Global financial market connectedness, as a percentage of variance explained



Notes: FCIs = financial conditions indices; GS = Goldman Sachs; IG = investment grade
Source: BIS 2025a.

SECTION III.

Implications and Policy Actions

The growth in nonbank financial intermediaries (NBFIs) and their associated vulnerabilities, and the increasing occurrence of stress associated with nonbank financial intermediation, suggest elevated risks of systemic events. Considering also the high financing needs of governments, the elevated economic and financial uncertainties, and growing geopolitical tensions and fragmentation, policy responses are urgently called for to prevent a systemic crisis from occurring and to better manage stresses. Policy steps needed include those that aim to reduce vulnerabilities and amplification mechanisms to lessen related systemic risks (preventive), and those that aim to respond better to diminish the impact of stress when it arises (remedial).

Many preventive actions have already been taken. The FSB and sectoral standard-setting bodies have been active in this area since the GFC, and even more so following the onset of Covid and its liquidity shocks.⁴³ Recommendations have been issued and adopted regarding money market funds, open-ended funds, and margining practices, and recently on monitoring nonbank leverage.⁴⁴ Implementation of these recommendations is critical, but they are incomplete. In addition, while central banks have made concerted efforts to address patterns of stress, they may be forced to come to the rescue again when stresses emerge.

But the rising financial stability risks from NBFIs call for additional actions on several fronts. Policy responses to address liquidity spillovers include steps to reduce vulnerabilities in core funding markets by limiting leverage; reduce the procyclicality of margining; increase the cost of bank financing to NBFIs; encourage securities market regulators to consider more financial stability risks; and develop better ways to provide liquidity in case of stress, including through international lending facilities and swap lines. To address solvency spillovers and enhance banking resilience, policymakers need to maintain limits and oversight on counterparty exposures; strictly supervise key financial institutions; conduct stress tests, notably of interconnections; and encourage greater burden sharing in recovery and resolution. To allow for better systemic risk analysis and improved market discipline, the quantity and quality of data on NBFIs need to increase and their availability to improve. Furthermore, monetary and fiscal policies need to internalize the risk taking and buildup of vulnerabilities in NBFI. When intervening, it is vital to clarify that monetary and financial stability policies have separate objectives and to communicate clearly when using monetary policy for financial stability.

Table 3 summarizes these key recommendations.

⁴³ FSI (2025); Knot (2025); and Schindler (2026) review actions and priorities.

⁴⁴ FSB 2025a.

TABLE 3: Policy Responses to Reduce and Manage NBFI-Related Systemic Risks

	Regulatory and Institutional		Other institutional and policy issues
	Liquidity spillovers to core bond and other funding markets	Solvency spillovers to main banking systems	
Preventive	• Have congruent regulations. • Conduct system-wide stress tests, analyzing potential liquidity spillovers. • Improve institutional infrastructure (e.g., CCPs). • Limit leverage by imposing minimum (possibly procyclically-adjusted) haircuts and initial margins. • Ensure all regulatory agencies consider and address systemic risks.	• Maintain counterparty and large exposure limits on banks. • Maintain strict supervision of banks, prime brokers, ICs, PFs. • Conduct system-wide stress tests, analyzing potential solvency spillovers. • Increase the cost of banks' securities financing to some NBFI-types.	• Improve data quality and quantity, ensure greater access by supervisory agencies, disclose more to enhance market discipline. • Monetary policy and fiscal policy to internalize the risk-taking and buildup of vulnerabilities in NBFIs.
Remedial	• Avoid open-ended asset purchases. • Ensure effective, broadly accessible (contingent or standing) international liquidity facilities and swap lines for major currencies. • Adopt (contingent) liquidity facilities for some NBFI-types with concurrent tight regulation and oversight.	• Greater burden sharing between the private and public sectors in recovery and resolution.	• Communicate clearly when using monetary tools for financial stability. • Deal with monetary and financial stability policy conflicts by careful design of respective programs and facilities.

Notes: green is continuation, i.e., maintaining existing policy and effort; yellow is intensification; red is innovation, i.e., introducing a new policy or action(s); CCPs = central clearing counterparties; ICs = insurance corporations; PFs = pension funds.

III.1. Address the risks of liquidity spillovers to core bond and other funding markets

Addressing the increasing NBF-related vulnerabilities in core markets. The starting point to reduce the risks of systemic events is to ensure that NBFIs and markets are well prepared to withstand high levels of stress on their own. The most reliable way to achieve this is through effective regulation and oversight, notably by reducing the amounts of liquidity transformation and leverage and by encouraging more resilient markets. This requires that market supervisors treat financial stability as a core element of their mandate to ensure market integrity, which in many jurisdictions may mean legislative or other institutional changes. That said, at times, policy-makers may have to provide tools and intervene to deal

with liquidity-driven stress, but that must come with costs and conditions for participants and limits on authorities.

Preventive regulatory actions. Efforts to reduce the risk and magnitude of liquidity imbalances mostly focus on reducing liquidity mismatches and constraining leverage. Besides those activity- and entity-based tools already being put in place,⁴⁵ additional policies can be used depending on context. A specific regulatory measure that could be effective in addressing liquidity- and leverage-related systemic risks in the repo markets is imposing minimum haircuts.⁴⁶ The near-zero haircuts today allow certain market participants to operate with high leverage using short-term financing, which creates fragilities. Minimum haircuts would reduce this but should be applied in a targeted manner to avoid inadvertently favoring one side of the trade—cash lenders or collateral

⁴⁵ FSB 2025b.

⁴⁶ FSB 2020.

lenders—while effectively addressing leverage buildup where it is most critical.

The procyclicality of margins requires attention given margins’ large liquidity demands at times.

Stress tests confirm the risks of large liquidity demands from margin calls.⁴⁷ A key trigger for margins’ procyclicality is the initial margins at central clearing counterparties and exchanges, which can be reset rapidly and discreetly in times of anticipated stresses, triggering asset price changes, with larger variation margins then to follow.⁴⁸ In addition to strict supervisory oversight and vetting of models, minimum (and possibly procyclically-adjusted) margins would help reduce these vulnerabilities.⁴⁹ As with haircuts, margin regulations are hard to implement and need to be granular enough to adequately recognize differences among the heterogeneous set of NBFIs and the extent to which they may contribute to systemic risks.⁵⁰

Actions should aim at reducing vulnerabilities that can lead to spillovers.

Other activity- and entity-based tools to be used should have “congruent regulation” as their guiding principle. This ensures that vulnerabilities are treated similarly across different types of financial institutions (including banks), while properly accounting for differences in business models and potential risks of amplification.⁵¹ For regulations to be consistent across activities and entities, a macroprudential policy approach might be needed.⁵² Entity rules can encourage diversification in the hedging assets used by securities firms issuing derivative-linked securities, to reduce the

high concentration today with risks of crowded trades, spillovers, and market dysfunction. Macroprudential regulation measures can strengthen the risk management of NBFIs, as was done in South Korea during the Covid-19 crisis. More generally, the private provision of (contingent) liquidity needs enhancement.

As a complementary measure, more resilient markets should be promoted.

Greater use of central clearing counterparties, more all-to-all trading systems,⁵³ and other improvements to the institutional infrastructure can help reduce systemic risks (box 1). This will also allow for better data and improved monitoring of liquidity mismatches and high leverage at NBFIs and the state of liquidity supply, including in foreign currency.

Better regulatory remedial tools for halting and managing liquidity stresses.

While financial intermediaries can use, and have used, various liquidity management tools, these do not always suffice and can have adverse market-wide effects (for example, due to adverse signaling). Regulatory steps have therefore been introduced to reduce systemic liquidity stresses and their severity. Greater use of circuit breakers, swing pricing, gates, and other “time-out limits” on convertibility actions triggered by specific market conditions or at the discretion of regulatory authorities could further help reduce systemic liquidity needs in case of shocks. Temporary exemptions from fair value accounting as well as using forms of risk pooling, that is, mutual insurance, can also possibly help overcome the externalities related to liquidity runs. At the same time, these and other such tools

47 For example, the 2024 Bank of England’s (BoE’s) system-wide exploratory scenario exercise found that approximately 85 percent of NBFIs’ liquidity needs would arise from variation margin calls, 8 percent from initial margin calls, and 7 percent from redemptions.

48 This relates to the models used for setting margins. (Although there can be review by supervisory agencies, the design of models is largely left to the industry.) Evidence (for example, Huang and Takáts 2020) suggests that the less capital a central clearing counterparty has at risk, the greater model-risk-sensitive its initial margins are to expected asset price changes (variation margins adjust to actual price changes). Also, margins for exchange-traded contracts typically adjust more than for bilaterally traded ones.

49 See FSB (2024c) for draft guidelines on enhancing the liquidity preparedness of nonbank market participants for margin and collateral calls in centrally and non-centrally-cleared derivatives and securities markets. This includes recommendations on conducting liquidity stress tests to identify sources of potential liquidity strains caused by margin and collateral calls, and to ensure a level of resilience consistent with their established liquidity risk appetite.

50 For example, in many repo transactions, haircuts are intended to protect not the cash lender but the collateral provider (that is, the cash borrower). A uniform application of minimum haircuts could inadvertently favor one side of the trade—cash lenders or collateral lenders—while failing to effectively address leverage buildup in the most critical areas.

51 See, for example, Metrick and Tarullo (2021) on achieving congruent regulation in the US Treasury market.

52 See Borio, Claessens, and Tarashev (2022) on balancing activity- and entity-based regulations.

53 All-to-all trading systems are electronic platforms allowing any market participant—including buy-side firms, dealers, and nonbank liquidity providers—to trade directly with one another, removing the need for traditional intermediary-based, bilateral relationships.

BOX 1: Better Institutional Infrastructures Can Help Reduce Systemic Risks

Complementary to regulatory reforms are improvements in and enhancements to institutional frameworks and infrastructures for core markets. Greater use of central clearing for cash and repo markets would enhance the resilience of government bond markets.^a Clearing requirements, which already cover various interest rate swaps, could be extended to FX swaps. More and better central clearing arrangements provide benefits in the form of greater netting and reduced counterparty and settlement risks. Central clearing also addresses the asymmetry present in bilateral markets, where haircuts for the largest hedge funds are smaller than for other participants due to market power. More netting of margins across instruments, including by making central clearing counterparties (CCPs) more interoperable, could also reduce systemic risks. Additional benefits could include better collateral management and general operational efficiencies. Finally, central clearing could free up balance sheet space for dealers (as they are only exposed to a clearinghouse), which boosts their intermediation capacity.^b Using guaranteed repos^c can free up balance sheet space, too, and reduce costs and improve operational efficiency.

Adopting all-to-all electronic platforms in segments that are still heavily dealer-dependent (for example, commercial paper, off-the-run sovereign bonds, corporate bonds) can also help reduce systemic risks. Often suggested for the US treasury market,^d such trading platforms

might improve market transparency and robustness and reduce the risk of their dysfunction. New technologies like distributed ledger technology might be useful here. Combined with increased central clearing, new trading platforms could also allow for a wider set of players with different trading motives, offering diversity benefits.^e That said, such changes could bring in additional agents (for example, algorithmic traders) contributing more opportunistically, further limiting liquidity in periods of stress.

More central clearing and all-to-all trading are not panaceas. Important challenges are the increased systemic risk of CCPs, notably when their “skin in the game” is low,^f and other critical nodes. Reforms can heighten the procyclicality of margins, as they tend to be higher in centralized markets than in bilateral trades. There are also unresolved questions around the scope of direct client access to CCPs. Moreover, while clearing allows for greater netting within a market, it can reduce netting across markets (for example, more netting within government bonds but less between government bonds and FX swaps). And it is not clear whether banks will use any freed-up balance sheet space to directly support bond markets or instead provide more brokerage services for hedge funds and others trading derivatives. While it may take more analysis, more central clearing, possibly combined with all-to-all trading, is likely to increase resilience.

Note: a. FSB 2022a. For the US Treasury market, see Duffie et al. (2023); Bowman, Huh, and Infante (2024); and Cenicola et al. (2026), with the latter finding that more central clearing would free up about \$34.5 billion in balance sheet capacity for the average US Global Systemically Important Bank (G-SIB). In addition, one could, as the US has done recently, exclude government securities from the calculation of banks’ (enhanced supplementary) leverage ratio, although this weakens the role of the leverage ratio as a backstop to risk-based capital adequacy requirements, potentially increasing systemic risks in the banking system. c. See Bloomberg 2025; and Sunthay.com. d. IAWG 2022; G30 2021, 2022; Liang and Parkinson 2020; Duffie 2022, 2023; Chaboud et al. 2022. e. For example, Duffie 2022. f. Cont and Ghamami 2025. g. For example, Singh 2026b.

need to be carefully designed to avoid triggering runs⁵⁴ and worsening regulatory arbitrage.⁵⁵

Liquidity support mechanisms need to adapt to address NBFI-related risks in core markets.

The main worry remains that spikes in NBFI-related liquidity demand can, through various spillover channels and forms of contagion, lead to even larger liquidity needs and have adverse overall effects. Dysfunctional core bond markets could in turn have global adverse effects. With commercial banks constrained in their ability and willingness to provide liquidity to NBFI, to fulfil their mandates central banks might occasionally again have to intervene to preserve market functioning and contain systemic

risks.⁵⁶ Providing liquidity support indirectly to banks that then pass it on to NBFI in need is often not effective, in large part because banks are reluctant in times of stress to on-lend such liquidity (this problem occurs irrespective of whether the central bank uses a supply-driven floor system, or a more demand-driven corridor system⁵⁷). To date, interventions have therefore mainly been in the form of large-scale or targeted central bank purchases of sovereign and other securities.

But there are many drawbacks and trade-offs to increasing public sector liquidity support to NBFI. As for other types of public support, such as the lender of last resort for banks,⁵⁸ the main drawback is the

54 For example, some have argued that first-mover advantages increased due to the post-GFC money market fund reforms in the US, adding to Covid-era runs.

55 Arbitrage can take many forms. Activities can move to other parts in the system or create spillovers, as shown in credit extension (for example, Cizel et al. 2019) and cross-border banking (for example, Claessens et al. 2023; Eren and Wooldridge 2022). See, also, Central Bank of Ireland (2023).

56 Markets Committee 2022; Buiter et al. 2023.

57 A supply-driven floor system maintains excess reserves, creating a “floor” for interest rates via administered rates, offering operational simplicity but requiring a larger central bank balance sheet. A corridor system supplies liquidity based on demand (or drains liquidity as necessary) to keep interest rates within a desired range, reducing the central bank footprint. See further Cavallino et al. 2025.

58 G30 2024.

expectation of support leading to more risk taking. This moral hazard can notably shift the underlying risk-reward calculus as it can rule out the worst possible losses, thus encouraging hedge funds and others in NBFIs to take on even more leverage.^{59,60} Furthermore, when formalizing support to NBFIs, one possibly unintended consequence could be weakening banks' franchise value. Unless access to liquidity support is accompanied by similar regulatory standards and prudential oversight that banks are subject to, it provides NBFIs a significant competitive advantage over banks and could increase, rather than decrease, systemic risk.⁶¹ In addition, emergency liquidity assistance comes with various implementation challenges and side effects.⁶² Liquidity support may conflict with other policies. Turmoil may arise precisely when high inflation calls for tighter monetary policy, raising complex messaging as to what is financial stability policy and what is monetary policy. In this context, the key overarching principles for interventions are that they should function as backstops, be temporary and targeted, use designs that mitigate side effects, and go hand in hand with effective regulation and oversight and the provision of key data.⁶³

Adopt central bank facilities to avoid large-scale purchases of sovereign bonds being the default case. Facilities can help reduce the risk of smaller stresses becoming systemic. If adopted, they can be activated in times of incipient stress or be standing facilities. For contingent emergency facilities, the central bank could announce clear conditions for when they may be activated. For all, it should be clear which NBFIs segments specifically have (or may have) access. Facilities'

access conditions can build on Bagehot's framework⁶⁴ used for 150 years for liquidity support to banks but adapted to balance the various trade-offs specific to the type of NBFIs.⁶⁵ The central bank should adopt and publish transparent eligibility rules that restrict access to lending or repos to solvent institutions and impose robust penalty rates and haircuts. Especially important is to have strong sunset and exit rules for facilities and publicly disclose how they were used once wound down.

NBFI-specific facilities have already been put in place by some central banks. One example is the BoE's recently adopted Contingent Non-Bank Financial Institution Repo Facility (CNRF), mainly for insurance corporations and pension funds. It is not a standing facility, but contingent on stress occurring, at which point its pricing will also be set. Reflecting the desire for selectivity and complementarity,⁶⁶ access and other criteria are established ex ante, adapting the Bagehot principles for lending to banks.⁶⁷ Proposals for providing (new) (collateralized) lending facilities for other classes of NBFIs also exist.⁶⁸ And some central banks are considering broadening the list of counterparties for their existing lending facilities to beyond commercial banks and further widening collateral pools. Central banks could also encourage indirect forms of liquidity support, for example, highly collateralized lending to commercial banks which, in turn, fund credit lines to liquidity-squeezed NBFIs,⁶⁹ and perhaps make these available even in normal times to ease incipient liquidity stress. The general challenge of these and other proposals, though, remains, which is containing moral hazard.

59 Shin 2025.

60 Moral hazard and its costs can vary by the specific type of assets supported and the nature of the shocks. For example, supporting a sovereign bond market that experienced stress following an exogenous shock may have less adverse moral hazard effects. And some of the gains of a central bank intervening at times could actually accrue to the public sector by making sovereign bonds more attractive to hold and use for various purposes. Nevertheless, it is most often not just the sovereign bonds that end up being supported, but also the trading strategies using such bonds.

61 It could also lead banks, invoking a level playing field, to call for less intense regulation and supervision and precipitate a race to the (regulatory) bottom (as described in Cetorelli [2025]).

62 Carstens 2021.

63 See, further, Markets Committee 2022.

64 Walter Bagehot, noted journalist, essayist, banker, and the second editor of *The Economist*, formulated his framework in *Lombard Street: A Description of the Money Market*, published in 1873. The framework established that during a panic, central banks should lend freely to solvent institutions against good collateral at a penalty rate. This can stop bank runs, reduce credit contraction, and avoid systemic collapse.

65 Acharya, Rajan, and Shu (2025) analyze how public provision of (contingent) liquidity can best improve the private provision of liquidity.

66 For example, Aramonte 2025.

67 Hauser 2023.

68 For example, Kashyap et al. (2025) for hedge funds. The pricing of these and other NBI-facilities could also involve a charge depending on the amount of liquidity support sought.

69 For example, Breckenfelder and Hoerova 2023.

In the extreme, the central bank could be the pawnbroker for all seasons. By extending the eligibility for their discount facility to any safe asset held by a solvent agent offering liquid liabilities, banks and non-banks, central banks could, in principle, mitigate almost all liquidity crises, as suggested by King and Tucker.⁷⁰ But there are serious doubts about whether this is the most efficient tool and whether engaging in broader support and acting as a general market maker of last resort does not create more moral hazard.⁷¹ To protect their credibility, central bankers must resist the temptation to treat every problem as systemic. Otherwise, the system will remain fragile by design, and the credibility of the public sector will erode every time it rescues another set of institutions that should have had more self-insurance or been allowed to fail. But being time- and incentive-consistent requires regulatory reforms to reduce risks.

Principles are needed, as real-life interventions show the many challenges and trade-offs to address. Principles should clarify how central banks intervene in general ways when a core market dysfunctions. Careful ex-ante design of contingent programs involving central bank bond purchases would allow for quick implementation in times of stress or crisis while minimizing negative side effects. If an asset purchase program is chosen, it might be better targeted, say, if risks in corporate bond markets dominate, than be wide, covering many classes of securities (to note, announcement effects may make actual purchases small). Programs for market functioning should incorporate penalty fees as well as conditions for exit and expiration. Even if the central bank buys on a full allotment basis, design choices for asset purchase still matter.⁷² BoE's actions in the recent UK liability-driven investment gilt crisis present a model. Notably, BoE's clear limits on the time window and quantity of support available helped make it a success. At the same time, it remained challenging to set the pricing and amounts of securities bought and to stick to a credible end date. And, as monetary policy was tightening, the intervention was not easy to communicate.

International liquidity support may be needed at times. The large cross-border dimensions of nonbank financial intermediation, especially through its significant reliance on FX swaps, can imply a need for international liquidity support. Typically, that support must be in a few selected currencies, notably the US dollar, given its dominance. For banks, standing swap lines among major central banks have been critical to stabilize the global financial system at times of acute distress.⁷³ But even then, liquidity in foreign currency is not always reliably available for banks, as was the case for Credit Suisse in March 2023. To date, such liquidity support is not ensured directly for NBFIs. This puts a premium on indirectly channeling foreign liquidity through banks to NBFIs, but this can be difficult, even domestically. As for other facilities, access will need to come with strict criteria set ex ante.

Some progress, but not easy or obviously sufficient, may be possible. Given greater fragmentation and less willingness to cooperate internationally, it may be difficult to keep existing swap lines in place (which may lead some foreign participants to withdraw from some bond markets). Still, some major central banks may want to extend swap lines to more central banks. Also, besides the European Central Bank, which recently broadened access to its repo facility (EUREP) to, in principle, all non-euro area central banks, others may want to modify their existing facilities or adopt new liquidity facilities. And the current US dollar repo facility for the official sector (the Foreign and International Monetary Authorities, FIMA) can possibly be expanded to longer maturities. Specific liquidity facilities with foreign domiciled NBFIs as counterparties appear less likely.⁷⁴ When swap lines and other liquidity facilities are less available or accessible than in the past, central banks from both emerging markets and advanced economies will have no choice but to gradually accumulate more FX reserves.

⁷⁰ King 2016, 2023; Tucker 2014, 2019.

⁷¹ G30 2024.

⁷² Duffie (2023) provides some suggestions.

⁷³ See further CGFS (2026) for cross-border funding risks and experiences with central bank foreign currency liquidity facilities.

⁷⁴ CGFS 2026.

III.2. Address the growing risks of solvency spillovers to banking systems

The greater role of NBFIs and their tighter interlinkages with banks increase the risks of adverse spillovers, with possible systemic effects. Both preventive and remedial policy responses are needed to protect the core of the banking system.

Banks themselves need, foremost, to manage the risks associated with interconnections.

In fact, addressing interconnections is part of normal banking business. This calls for good internal governance that ensures proper bank risk management.⁷⁵ However, not all banks manage risks well, as past and recent episodes⁷⁶ have shown. Idiosyncratic risks, including related to the two-way connections with NBFIs, may arise. When it can become systemic, public sector actions are called for.

Proper and consistent microprudential regulations and supervisory vigilance should be reinforced.

As risks are increasing, this is not the time for jurisdictions to relax regulations and further fragment the global regulatory framework. Rather, as unanimously expressed in March 2026 by the Group of Central Bank Governors and Heads of Supervision: “GHOS members welcomed the implementation progress and reaffirmed their expectation of full and consistent implementation of the Basel III framework by all [Basel Committee on Banking Supervision, BCBS] member jurisdictions as soon as possible.”⁷⁷ While there may be some room to streamline the application of certain regulations, this should reduce, not increase, the scope for spillovers through links between banks and NBFIs. Nor should supervisory attention, to these connections or in general, be lowered. Rather it could be necessary to intensify

oversight, notably of banks’ fast-growing links through prime brokerage and private credit financing.

Reinforcement might include new regulatory measures.

Proper regulation and supervision will help mitigate a buildup of vulnerabilities. But competition puts pressure on banks to keep supporting private credit, hedge funds, and other NBFIs (too cheaply), even in the face of repeated mistakes and increasing signs of vulnerabilities. So, collective action problems remain. As links between the sectors evolve, distress could also manifest or spread in unexpected ways. This can call for new steps. Regulations on banks can be an indirect way to reduce NBFI risks but need to be balanced against the risks of encouraging more migration of activities from banks to NBFIs. An early suggestion was a security financing charge, which makes banks price their financing of NBFI (via their prime-brokerage services, typically in the form of a repo) considering its overall systemic risk.⁷⁸ To meet its macroprudential objectives, such a charge could be adjusted procyclically or set at a fixed level (recent experience with the countercyclical bank buffer and other analyses suggests a fixed level may suffice).⁷⁹

Step up stress tests of banks and other NBFI-related risks.

For jurisdictions where NBFI is sizeable and macroeconomically and systemically critical, routine bank stress tests should assess banks’ links to various NBFI segments. Links can transfer value through exploiting regulatory arbitrage opportunities and because banks are ultimately backstopped. And links can be avenues for spillovers. Supervisors already routinely collect for banks data on large counterparties and exposures. Tests should cover both on- and off-balance-sheet items and analyze for which NBFI entities banks, individually and collectively, have important to-and- from obligations, and through what instruments. This way, banks, NBFI segments, and institutions’ potential key to NBFI-related systemic stress,

⁷⁵ G30 2012; 2013; 2015; 2018.

⁷⁶ G30 2024.

⁷⁷ BCBS (2026).

⁷⁸ Tarullo 2012; see also Andersen, Duffie, and Song 2019.

⁷⁹ Aldasoro, Huang, and Tarashev (2021) show that by imposing a permanent floor on the cost of liquidity provision (e.g., through a repo), bank regulation disciplines risk taking in the asset management sector, raising social welfare. They suggest using leverage-ratio regulation, for which they find a risk-reducing effect on money market funds. (Leverage-ratio regulation acts as a capital backstop for banks, requiring them to hold a minimum amount of Tier 1 (highest quality) capital against total assets and off-balance-sheet exposures. Established under Basel III, it prevents excessive borrowing, ensuring banks can absorb losses regardless of asset risk.)

can be identified at relatively low effort. Also, effects of potential shocks to and from NBFI, their amplification, and effects on financial markets, including core markets, can be analyzed. Several agencies already do this regularly⁸⁰ or as necessary.⁸¹ In case systemic vulnerabilities are identified this way, (individual) bank's capital and liquidity requirements could be adjusted to rein them in and boost resilience, again as ways to internalize some of the financial and real sector externalities neither banks nor NBFIs price.

Review and analyze other leverage-related risks. Leverage-related risks are not always of a systemic nature. For many NBFI types, leverage-related losses are shared (over time) with their stakeholders, making systemic consequences less likely. Regardless, monitoring leverage-related financial stability risks in NBFI is important.⁸² Following those agencies that already regularly do so, generic stress tests should be used for all institutions that are large and where leverage, either direct or synthetic, can lead to adverse spillovers. The type of institutions covered will vary by country but typically include pension funds, insurance corporations, hedge funds, and possibly other leveraged funds, and, where relevant, banks and broker-dealers in their role as leverage providers to NBFIs. Preferably, given NBFIs' footprint, such stress tests should also be done at the global level by key supervisory agencies cooperating on common stress scenarios and sharing data and analyses.

Analyze systemic liquidity risks arising from interconnections. Supervisory agencies in major financial systems already routinely perform stress tests as to how investment fund liquidity needs could spill over system-wide, and some review their central clearing counterparties for counterparty credit risk. Liquidity spillovers from other NBFI types, often related to margin and collateral calls, have become more important and need to be analyzed, as well. Transaction-level data, typically available to some supervisory agencies, can be used. Already, the Banco Central do Brasil, BoE, and European

Securities and Markets Authority (ESMA) conduct such reviews, and they also regularly stress test their central clearing counterparties.⁸³ The goal is to obtain an overall perspective on the key subsector(s) in nonbank financial intermediation that could present systemic risk. Exercises like BoE's 2024 system-wide exploratory scenario exercise can provide system-wide perspectives of how market shocks can lead to liquidity shortfalls and related spillovers. Broader spillover analyses could assess risks emerging from other links, as when shocks to international financial conditions are amplified because of NBFI-related linkages (for example, due to capital outflows).

Remedially, more burden sharing is possible.

The typical response to banking stress is unfortunately to invoke the safety net, either in the form of system-wide lending facilities, or higher deposit insurance and blanket guarantees, which come with moral hazard and other distortions. Similarly, support for NBFI-related stresses has its costs. To improve incentives, the public safety net should be less generous in quantity and higher in costs. More burden sharing is especially feasible for NBFIs as their individual activities and entities are more geared toward risk sharing. Improvements can come from having shareholders and creditors incur more of the cost of default. Using more mutual structures (for example, sharing the cost of an NBFI recovery or resolution) might also increase market discipline and help offset the private benefits some institutions receive. But such recommendations hinge on the ability to contain adverse spillovers, which calls for credible means in place to potentially provide support to other sound institutions.

III.3. Enhance NBFI's transparency by increasing data and their availability and disclosure

With more NBFI, financial stability risks have become more opaque and harder to monitor. Some NBFI parts are quite transparent (for example,

80 For example, European Banking Authority; see Barberi et al. (2025).

81 For example, BoE 2025.

82 FSB (2024a) provides recommendations.

83 For the US, Heilbron and Tompaidis (2025) model shocks for 11 central clearing counterparties (CCPs) and six large clearing members and stress CCPs in several ways.

investment funds often disclose at the security level and at high frequency to the public and supervisory agencies). But data on many other key parts of NBFIs can be sparse and poorly disclosed. Better data on only balance sheets does not suffice as many instruments are off-balance-sheet (for example, synthetic risk transfers). Compared to banks, the complexity of NBFIs can also be much greater, with numerous types of entities, activities, and financial instruments. Relatedly, the opaqueness in many parts of NBFIs is high, for example, there is a great reliance on private ratings (that is, those not from public rating agencies). Notably, the growing interlinkages among NBFIs and between NBFIs and banks are hard to gauge by financial markets and supervisory agencies. For example, the aggregate exposures of Archegos were not visible to the banks themselves (or to any supervisory agencies for that matter), let alone disclosed. As in earlier cases (for example, Long-Term Capital Management), this led many banks to

underestimate counterparty risk. More generally, lack of good data means more known and unknown unknowns for authorities and markets, with associated risks of missed vulnerabilities and unexpected behaviors.

There is, consequently, a need to improve the amount and quality of data about nonbank financial intermediation, access to data, and disclosure of key data. Box 2 summarizes deficiencies and ongoing work on improving data. Some gaps can be filled using existing data through better access and sharing among authorities. Other gaps, particularly in interlinkages, cross-sector, and cross-border, need new or better reporting requirements and data collection. This will require some carrots and sticks to get key players (for example, hedge funds, high-frequency traders) to provide data. And any new regulation and access to liquidity facilities must come with data reporting duties.

BOX 2: Data on NBFIs Entities and Activities: Ongoing and Further Improvements

More and better nonbank financial intermediary-related data are being collected. At the national and international level, more data are being collected and becoming more accessible to supervisory agencies and publicly. This allows for better NBFIs stress tests and analysis of potential vulnerabilities and the impacts of (new) policy measures. Progress also comes from advances in data collection and management. Further gains are starting to be made by using machine learning, artificial intelligence (AI), and related tools to collect and process data, including from publicly available sources (for example, texts and text-based filings) and from nontraditional sources.

But many important gaps remain. As the Financial Stability Board/International Monetary Fund Data Gaps Initiative (see IMF latest update for current status), FSB (2025c), and others document, many data gaps remain for NBFIs at the aggregate and subaggregate levels (for example, by type of entity or activity) and at the single entity level. For segments like hedge funds, Principal Trading Firms, and private credit, too little data are collected. While volumes of pledged collateral, an important part of NBFIs funding to banks and a form of leverage, can be pieced together from bank disclosures (in footnotes to annual reports and the like of Global Systemically Important Bank [G-SIBs], often requiring verification with relevant banks), they are not available in a single database.^a Data on final NBFIs investors are often missing, and NBFIs' significant intrasectoral and cross-border activities continue to complicate, in large part as standards vary and privacy and supervisory confidentiality rules limit data sharing. Other activities have large gaps, as well.^b Some of these challenges will increase with tokenization and new forms of financial services due to digitalization.^c

Efforts continuously need prioritization. By focusing exclusively on filling data gaps, resources can be spent ineffectively. Clear framing helps

to collect what is most important, what is doable, and how to best reap efficiency gains. This also applies to how to best connect various existing data so as to best capture exposures and interactions that can lead to stress.^d For interconnections, bank supervisory data can be used more to identify key asset exposures to NBFIs, but data on funding often still lack details on NBFIs-bank links. For liquidity stress tests, data from monetary policy operations can help to better identify common pressure points in funding markets. For capital and derivatives markets, key propagators, positions of high-frequency traders, hedge funds, and others that involve (implicit) leverage and liquidity transformation and crowded trades in core markets can be better documented.

Important opportunities exist. Data from trading repositories and securities settlement systems can be used as these typically already report to supervisory agencies on order books, actual margins, and haircuts.^e Using (price and quantity) data from data providers commonly used by banks and NBFIs presents further opportunities. With regard to margining, gaps can partly be addressed by enhancing market disclosure, but separate reporting processes will be needed for those (bilateral) transactions not suited to disclosure.

Better standards to enhance data quality and greater disclosures are needed. Data obviously must be as comparable as possible and shareable among relevant authorities within and across jurisdictions. Standards thus need to be harmonized for data recording and matching (for example, using a Legal Entity Identifier for the identification of various links). Reporting requirements should reflect progress in vulnerability metrics and incorporate analytical-based measures (for example, besides leverage and detailed balance sheets, also measures on NBFIs' interactions with asset prices and other market characteristics). Not all (existing and new) data need to become publicly available, but key data should be,

and reporting frequency can increase. This will help the private sector better manage risks (for example, counterparty concentration), allow for more market discipline and better systemic risk analyses, and can help avoid stress events (for example, as better information stops agents from rushing to the door first).

The ability to collect and share (in principle) relevant data nationally and internationally needs to be improved. In some jurisdictions, security market supervisors do not have the right to request (real-time) data from intermediaries. Even when collected, data are often not accessible in a timely and easily usable manner, and differences in regulatory coverage create legal limits for accessing data

and sharing them among domestic authorities, including those charged with systemic risk analysis.^f Limitations on data sharing across borders are often compounded by, besides legal obligations, technical impediments hindering effective data sharing.^g

Explore further synergies with regulatory reforms and private sector efforts. Better data will in part be driven by key regulatory reforms, including those discussed internationally, to limit NBF-related systemic risks. When accompanied by reporting and disclosures, they can improve data. The private sector's interests can be better aligned through partnerships with the industry to help foster improved standards and reporting guidelines. All this can help avoid the repeat of stress events.

Note: a. Singh and Alam 2018. b. For example, FSB (2023a) on open-ended fund liquidity imbalances; FSB (2024a) on margining; FSB (2025a) on leveraged trading strategies in core markets, its current test case for addressing data challenges; and FSB (2026) on repo markets. c. For example, Singh 2026a. d. Leverage and liquidity imbalances thought to be "hidden" are, ex post, often found to have been discoverable with ex-ante, already available data, using, of course, sufficient resources. e. CPMI-IOSCO 2013; Heitfield 2014. f. Many NBF transactions are channeled through central clearing counterparties and other Financial Market Infrastructures (FMIs). The resulting trade repository data cover key position and interconnection aspects useful to analyze the resilience of markets, FMIs, and agents, but data often cannot be shared domestically, let alone internationally. g. See, further, FSB 2025c.

III.4. Internalize NBF systemic risks more in monetary and fiscal policies

Monetary policy must internalize, to a greater extent, the risk-taking and buildup of vulnerabilities in NBF and clarify its communication.

The separation principle between monetary and financial stability policy does not always hold.⁸⁴ Monetary policy must, to a greater extent, consider the risk-taking in the NBF parts of the financial system where it is less fully controlled by regulation and supervision. Furthermore, monetary policy must consider its asymmetric effects on NBFs, which can differ from those for banks.⁸⁵ When tightening, vulnerabilities can imply limits on policy, as revealed for banks in March 2023,⁸⁶ but also for NBFs, as in the 2022 UK liability-driven investment crisis. Financial stability interventions and related communications become even more challenging when tightening. Dealing with conflicting policy objectives can partly be

addressed by careful design of respective programs and facilities.⁸⁷ But policymakers will also need to distinguish clearly in their communications between quantitative easing for market functioning versus monetary stimulus.

Some fiscal policy recommendations relate.

Fiscal policy should also consider the possible consequences of the risk taking and vulnerabilities in NBF, notably by helping preserve the headroom that may be needed to deal with systemic crises. And fiscal consolidation in major economies is, regardless, needed for a variety of reasons.^{88,89} Other, NBF-specific fiscal policy recommendations are more limited. Regardless, any adjustment to fiscal policy, public debt management, or other policies related to NBF developments should not undermine fiscal soundness, increase sovereign rollover and interest risks, or lead to greater sovereign-financial system links, as well-managed public finance is the bedrock of financial and monetary stability.

⁸⁴ G30 2023; 2024; 2025b.

⁸⁵ In contrast to banks holding long-dated securities, higher interest rates typically improve the solvency position of pension funds and life insurance corporations, as the duration of their liabilities exceeds that of their assets. This may explain why higher interest rates lead some NBFs to extend more financing. Also, NBFs relying more on long-term investors can be less sensitive to rate increases. Other NBFs, in contrast, may react more like banks and reduce their credit extension when interest rates rise. See Claessens (2025) and Gelos (2025) for initial reviews on monetary policy transmission in the context of greater nonbank financial intermediation and how it varies for specific NBF types.

⁸⁶ G30 2024.

⁸⁷ G30 2025b.

⁸⁸ IMF 2025c.

⁸⁹ Notably, rising government debt could risk fiscal dominance (Carstens 2025; Cochrane 2023).

SECTION IV.

Areas that Need Further Analysis and Possible Action

The proposals made here have implications for various agents. Much needs to be done by authorities to prioritize the various possible preventive actions by conducting further analyses on how best to calibrate them, address their potential adverse side effects, and putting systems, including data for monitoring, in place. Supervisors will need to conduct more system-wide stress tests to identify NBFI-related vulnerabilities. As for most regulation, analyses will have to balance reducing systemic risks and ensuring that the economic and financial needs for NBFIs's services are adequately met. In all of this, it is necessary to maintain an international level playing field to avoid the migration of NBFI activities to jurisdictions with the least stringent regulation and oversight.

By being preemptive now, rather than easing regulations, systemic vulnerabilities can materially be lessened. But stresses may still emerge. Central banks, especially those with a large NBFI sector, will need to update their policies and systems for possibly extending liquidity to a new set of entities and activities. This, too, will require significant analysis as well as investments in institutional infrastructures and systems. It needs to address questions like how to design facilities to retain as much market discipline as possible, how to ensure that entities with potential access are adequately regulated and overseen, and what the requirements are in terms of greater transparency and provision of data. Central banks also need to review how monetary policy and its communication may need to change in light of any NBFI-related risk taking and buildup of vulnerabilities and stability interventions.

A specific suggestion is to conduct more NBFI-related crisis simulation exercises. Traditionally done for banking systems, such exercises can be adapted to test the preparedness of financial authorities, regulators, supervisory agencies, and institutions for NBFI-related crises. Exercises would inform the robustness of decision-making, communication, national and cross-border coordination, and legal frameworks, notably under time pressure. They could also speak to the adequacy of liquidity support facilities and their designs. They can enhance the collective preparedness to a possible systemic NBFI-related event and inform potential gaps in preventive measures, especially when done as a collaborative cross-country exercise and covering the most important financial institutions worldwide, as has been done for banks and central clearing counterparties.

Assessing the impact of new developments. Besides these regulatory, supervisory, and central bank actions, there are areas where more data should be collected, developments monitored more closely as to the buildup of vulnerabilities, and the possibility of regulatory and oversight actions assessed to prevent systemic risks from emerging. These areas include new forms of NBFIs, the roles and impacts of AI, and more generally the ongoing digitalization.

New forms of nonbank financial intermediation. Reflecting its dynamism, nonbank financial intermediation is ever evolving and often at the forefront

of financial innovation. This includes incorporating technological advances, as shown by three new intermediation forms that have recently emerged: cryptocurrencies, decentralized finance (DeFi), and stablecoins.⁹⁰ While still small in market size, as they grow and integrate with the main financial system, they could raise the types of systemic risks often seen with financial innovation.

Systemic vulnerabilities appear most imminent for stablecoins.⁹¹ The total value of stablecoins is currently only about \$300 billion. But helped by recently adopted regulatory frameworks, such as those of the EU, the US, and others⁹² along with detailed guidance from supervisory agencies, substantial new issuance is expected (some forecast the total value of stablecoins outstanding to reach more than \$2 trillion over the next five years.⁹³ Globally, however, the regulatory landscape still varies.⁹⁴ And questions are being raised about whether the rules being adopted and accompanying supervision foreseen are comprehensive enough to prevent the emergence of systemic risks.^{95, 96} Weak rules and limited oversight could lead to a repeat of earlier experiences for banks, with widespread runs and a lack of monetary singleness,⁹⁷ problems that only subsequently were addressed.^{98, 99} Furthermore, when (largely) holding short-term sovereign debt, sovereign-stablecoin links intensify, with risk of market dysfunction in case of stablecoin runs. Major risks, and not only for emerging markets and developing economies, are forms of currency flight and loss of monetary sovereignty,¹⁰⁰ in part as using stablecoins can more easily circumvent capital controls.¹⁰¹

Artificial intelligence is greatly affecting NBFIs, potentially leading to systemic risks.

AI is affecting financial systems and authorities worldwide.¹⁰² Being at the frontier of technology, various NBFIs (for example, high-frequency traders) have for some time combined large datasets with machine learning and other such techniques. But all kinds of AI use are spreading rapidly across the broader NBFIs space. If used properly, this can improve the efficiency of financial services provision and expand the quality and mix of products offered, leading to productivity gains in the real economy. AI can also enhance entities' risk management and lead to better monitoring by markets and supervisors.

At the same, AI can potentially create or add to NBFIs-related systemic risk, in sometimes known, but often unknown, ways. It is likely to add to operational and concentration risks (for example, given greater third-party dependencies). In addition, it could mean more volatility, increase the speed of herding and contagion, and lead to collusion as AI models are similar or start to react to each other in perverse ways.¹⁰³ It could lead to market structure changes, which could reduce incentives to monitor agents. Other effects of AI, many still unforeseen, could increase risks. With little public action to date, the possibility of systemic events grows. As such, monitoring the various AI developments in the financial services industries and their impacts, and learning what AI may mean for financial stability, are now important actions for supervisory agencies and central banks alike.

90 Aquilina et al. (2025a); Aldasoro et al. (2025); and BIS (2018, 2022, 2025b) analyze aspects of their emergence, size, and risks.

91 Risks for DeFi and cryptocurrencies are mostly that individual defaults and fraud trigger runs and have adverse confidence effects, which can spill over to mainstream finance.

92 The EU, with its Markets in Crypto-Assets (MiCA) Regulation; the US, with its Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act); Singapore, with its Stablecoin Regulatory Framework; Japan with its Payment Services Act (amendment); and Canada, Switzerland and the UK with their (upcoming) regulatory regimes.

93 For example, see Citibank 2025; and Adrian et al. 2025, footnote 22.

94 FSB 2025b.

95 Ahmed et al. 2025; ESRB 2025; Gross and Stenner 2026; Liang and Dudley 2026.

96 Notably, in some jurisdictions the type of assets stablecoins can hold as backing for their liabilities is less restrictive than for banks and investment funds, while run risks can be similar. Vehicles prone to run risks are traditionally also more tightly overseen than some stablecoins appear to become.

97 The BIS (2025b, page 78) defines monetary singleness as follows: "Common knowledge of the value of money has a shorthand – the "singleness of money" – where money can be issued by different banks and accepted by all without hesitation."

98 G30 2025b.

99 Accordingly, there are questions on what stablecoins mean for monetary policy (for example, BIS 2025b).

100 IMF 2021.

101 Cerutti, Chen, and Hengge 2024.

102 IMF 2024b; FSB 2025f; BIS 2024.

103 For example, IMF 2024b.

Greater digitalization is introducing new risks, including within NBFIs. Beyond the emergence of new forms of money and decentralized architectures for financial service provision supported by distributed ledger technology, digitalization is reshaping financial services more broadly. It is enabling faster runs, leading to greater contagion, and making for more highly correlated price

movements. Digitalization may also alter market structures, weaken monitoring incentives, and undermine traditional mechanisms for stabilizing markets. Together, these developments call for further analysis and may require adjustments to regulatory frameworks beyond those addressed in this report.

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